

Supplementary information

Tools for the shifted-zeta Wilson–Loewner program

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21 September 2026

Live revision of version 0.5 (not archived)

Abstract

This companion develops the technical calculations behind the physical Wilson–Loewner program and its longer-term arithmetic realization question. For a genuine Loewner prefix completed by a return chord in fixed four-dimensional Yang–Mills theory, we prove the moving-tip cancellation, derive the first two insertion equations, and establish a short-time expectation expansion with explicit smooth-field error bounds. We retain the curvature ordering, transverse-current residual and field-flow response needed for a Schwinger–Dyson closure attempt. Quantum averaging requires stated regularity and moment assumptions; no continuum construction or closed scalar loop equation is asserted. Earlier sections preserve the causal arithmetic factorization, free defect kernel, smooth contour variation, endpoint selection, reflected pairing and Gaussian response controls. They also give the fixed-window readout obstructions, auxiliary boson–fermion product, localized hemisphere amplitude, radial non-descent and canonical-Hodge comparison. Arithmetic anchor and append results are reported as separate internal work. The original 299 diagnostics and additional localization controls retain their own records; new trace-growth checks test geometry and ordered matrix calculus, not quantum expectation values. None of these results supplies the physical arithmetic transfer or its positive cumulative norm.

How to read this companion. The program, its physical realization hypothesis and the route to RH are developed in the main exposition [1]. Sections S1–S8 retain the technical calculations of the preceding unified manuscript, with arithmetic conventions repeated for independent reading. Section S9 records the inherited continuation algebra; Sections S10–S11 give the fixed-window response test and localization product. Section S12 gives the predictive localization controls, radial non-descent and Hodge alternative. Section S13 gives the genuine-trace quantum hierarchy, its remainder bound and the specific closure residuals; Sections S14–S15 give diagnostics and provenance. Section S9 records the separate arithmetic anchor, scalar-coupling obstruction and subsequent operator-coupling result. The two documents have separate numbering and bibliographies and can be compiled independently. References to numbered sections below are local unless explicitly attributed to the exposition.

Contents

S1	Arithmetic conventions and exact kernel factorization	3
S1.1	The compactly supported Weil form	3
S1.2	The positive kinetic term and the indispensable local term	4
S1.3	Shifted transfers and a sufficient contraction route	5
S1.4	The exact positive part, and what it leaves unresolved	6

S2	Why defect matter reproduces the archimedean exponents	7
S2.1	An auxiliary Gaussian realization of the gamma ratio	8
S3	Semicircles, constant drivers, and a restricted rigidity theorem	8
S3.1	Fixed scalar coupling and fixed special charge	9
S4	Smooth Wilson variation and the Loewner insertion	10
S4.1	Conventions and the exact variation	10
S4.2	Regular inverse-map deformations	11
S4.3	A precise obstruction to geometry-only closure	11
S4.4	The tangent connection as a one-form	12
S5	Tangent coupling with the full chiral defect projector	12
S5.1	Field content and conventions	12
S5.2	A normal plane	13
S5.3	One charge across all defect directions	13
S5.4	The remaining H weight	14
S6	Endpoint polarization, vanishing, and the adjoint	14
S6.1	The same-charge endpoint calculation	14
S6.2	The full bilinear has nonzero R charge	15
S6.3	Why the original semicircle is a nonzero control	15
S6.4	The adjoint changes the scalar map	16
S7	Physical reflection and the reference color junction	16
S7.1	Ordinary time reflection	16
S7.2	The actual paired operator and the conditional Gram statement	16
S7.3	An unconditional free kernel	17
S7.4	The two branches preserve different constant charges	17
S7.5	Internal rotations do not automatically preserve positivity	18
S7.6	A free test of an odd-H normal reflection	18
S8	A smooth reference join and a direct bulk response	19
S8.1	A contour family that meets the reflection plane smoothly	19
S8.2	The specified Gaussian perturbative sector	20
S8.3	Why a positive norm need not be protected	21
S8.4	The complete response still required	21
S9	Cumulative-storage gluing and joint continuation	22
S9.1	The fixed-shift spatial Schur complement	22
S9.2	Cayley coordinates and reflection	22
S9.3	A step changing both length and shift	23
S9.4	Scope near zero shift and at increasing depth	24
S9.5	Separate arithmetic anchor and append results	24
S10	A normalized Wilson response on a fixed window	25
S10.1	The causal target and its fixed contact term	25
S10.2	The lowest-mode cancellation at finite shift	26
S10.3	A regulated readout with identity initial data	27

S10.4	A nonzero but bounded Gaussian first variation	27
S10.5	Two obstructions and their precise scope	28
S10.6	Consequences for storage and continuation	29
S11	Localization and an auxiliary boson–fermion product	29
S11.1	The defect-compatible localization question	30
S11.2	An exact product for the complete prime-free factor	30
S11.3	A mode-counting test and the remaining dictionary	31
S12	Localized boundary amplitudes and spatial radial evolution	31
S12.1	A specified charge, boundary state and circle control	32
S12.2	The actual two-scalar state and its adjoint	33
S12.3	Why ordinary spatial dilation fails on fixed representatives	33
S12.4	Canonical Hodge evolution and the lost spatial tower	34
S12.5	Composite-state characters and endpoint spectral weights	35
S12.6	Scope of the arithmetic comparison	35
S13	Genuine trace growth in Yang–Mills theory	36
S13.1	Theory, field resolution and averaging assumptions	36
S13.2	The actual prefix and its return chord	37
S13.3	Exact cancellation of the moving-tip term	37
S13.4	Curvature transport and the next expectation equation	38
S13.5	The explicit trace of a linear driver	38
S13.6	A quantitative short-time expectation bound	39
S13.7	Schwinger–Dyson equations and the unresolved closure	40
S14	Reproducible diagnostics and controls	41
S14.1	Additional localization controls in version 0.5	42
S14.2	Genuine trace-growth controls in the live revision	42
S15	Provenance and contribution disclosure	43

S1 Arithmetic conventions and exact kernel factorization

We repeat the arithmetic normalization to make this companion independently readable. The effective-evolution hypothesis and its role in the program are stated in the exposition [1]. The free defect calculations below use the exponential tower

$$n_{\Gamma}(u) = \frac{e^{-u/2}}{1 - e^{-2u}} = \sum_{n=0}^{\infty} e^{-(2n+1/2)u}, \quad u > 0. \quad (\text{S1.1})$$

S1.1 The compactly supported Weil form

Let $I_L = (-L/2, L/2)$, with $L > 0$, and let $f \in C_c^{\infty}(I_L; \mathbb{C})$. Write F for its zero extension to $\mathbb{R}$ and use

$$\hat{F}(\tau) = \int_{\mathbb{R}} e^{-i\tau x} F(x) \, dx, \quad \langle f, g \rangle = \int_{I_L} \overline{f(x)} g(x) \, dx. \quad (\text{S1.2})$$

For $a > 0$, define the compressed shift $T_a f(x) = F(x - a)$ on I_L . Let $\psi = \Gamma'/\Gamma$ be the digamma function, and let $\Lambda(n) = \log p$ when $n = p^k$, p prime and $k \geq 1$, and zero otherwise. Finally put

$$C(f) = \int_{I_L} f(x) \cosh(x/2) dx, \quad S(f) = \int_{I_L} f(x) \sinh(x/2) dx. \quad (\text{S1.3})$$

The normalization of the program is

$$\begin{aligned} Q_{0,L}[f] &= \frac{1}{2\pi} \int_{\mathbb{R}} [\operatorname{Re} \psi(\tfrac{1}{4} + \tfrac{i\tau}{2}) - \log \pi] |\widehat{F}(\tau)|^2 d\tau + 2|C(f)|^2 - 2|S(f)|^2 \\ &\quad - \sum_{\substack{n \geq 2 \\ \log n < L}} \frac{\Lambda(n)}{\sqrt{n}} \langle f, (T_{\log n} + T_{\log n}^*) f \rangle. \end{aligned} \quad (\text{S1.4})$$

The prime-power sum is finite on a fixed interval. The pole term is indefinite, and the archimedean multiplier includes its local constant. Neither may be dropped in a positivity argument.

For comparison with the usual explicit formula, set $h = F * \tilde{F}$, where $\tilde{F}(x) = \overline{F(-x)}$. The two prime sums in the Weil functional give the last line of (S1.4); its $e^{x/2} + e^{-x/2}$ term gives $2|C|^2 - 2|S|^2$. Its archimedean distribution has Fourier multiplier $\operatorname{Re} \psi(1/4 + i\tau/2) - \log \pi$. Thus $Q_{0,L}[f] = W(F * \tilde{F})$ in the convention of [11, p. 1]. In particular the weight is $\Lambda(n)/\sqrt{n}$.

Theorem S1.1 (Weil positivity criterion, used as background). *RH holds if and only if $Q_{0,L}[f] \geq 0$ for every $L > 0$ and every $f \in C_c^\infty(I_L; \mathbb{C})$.*

The equivalence is the classical criterion, not a result of the present construction. Every compact support lies in one such interval, so an unbounded sequence of interval lengths suffices. A finite collection of lengths or numerical matrices does not.

S1.2 The positive kinetic term and the indispensable local term

Put $a_n = 2n + 1/2$, for $n \geq 0$, and

$$b(\tau^2) = \operatorname{Re} \psi(\tfrac{1}{4} + \tfrac{i\tau}{2}) - \psi(\tfrac{1}{4}) = 2 \sum_{n=0}^{\infty} \frac{\tau^2}{a_n(a_n^2 + \tau^2)} \geq 0, \quad w_0 = \psi(\tfrac{1}{4}) - \log \pi < 0. \quad (\text{S1.5})$$

The constant $\psi(1/4)$ is not equal to $\log \pi$. Adding and subtracting $\psi(1/4)$ in the original multiplier gives the exact identity

$$\begin{aligned} \operatorname{Re} \psi(\tfrac{1}{4} + \tfrac{i\tau}{2}) - \log \pi &= [\operatorname{Re} \psi(\tfrac{1}{4} + \tfrac{i\tau}{2}) - \psi(\tfrac{1}{4})] + [\psi(\tfrac{1}{4}) - \log \pi] \\ &= b(\tau^2) + w_0. \end{aligned} \quad (\text{S1.6})$$

Thus w_0 is the remainder after separating the nonnegative multiplier $b(\tau^2)$; it is already present in (S1.4), not an additional term introduced into the form.

The series in (S1.5) follows by subtracting the convergent digamma partial fractions. Each exponential in (S1.1) satisfies $\int_0^\infty e^{-au} (1 - \cos \tau u) du = \tau^2/[a(a^2 + \tau^2)]$. Plancherel therefore gives

$$\mathcal{E}_\Gamma[F] := \int_0^\infty n_\Gamma(u) \|F(\cdot + u) - F\|_2^2 du = \frac{1}{2\pi} \int_{\mathbb{R}} b(\tau^2) |\widehat{F}(\tau)|^2 d\tau. \quad (\text{S1.7})$$

Using (S1.6) and $\frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{F}(\tau)|^2 d\tau = \|f\|_2^2$, the archimedean contribution becomes

$$\frac{1}{2\pi} \int_{\mathbb{R}} [\operatorname{Re} \psi(\tfrac{1}{4} + \tfrac{i\tau}{2}) - \log \pi] |\widehat{F}(\tau)|^2 d\tau = \mathcal{E}_\Gamma[F] + w_0 \|f\|_2^2. \quad (\text{S1.8})$$

Substituting this identity into (S1.4) rewrites the full target as

$$Q_{0,L}[f] = \mathcal{E}_\Gamma[F] + w_0 \|f\|_2^2 + 2|C(f)|^2 - 2|S(f)|^2 - \sum_{\substack{n \geq 2 \\ \log n < L}} \frac{\Lambda(n)}{\sqrt{n}} \langle f, (T_{\log n} + T_{\log n}^*)f \rangle. \quad (\text{S1.9})$$

Here $w_0 \|f\|_2^2 = w_0 \int_{I_L} |f(x)|^2 dx$ is the fixed negative local term. The positive difference energy is only one part of the desired form: its local term, pole contribution and prime-power translations must all be retained.

The natural form domain on a fixed interval is given by zero extensions with $\int \log(2 + |\tau|) |\widehat{F}(\tau)|^2 d\tau < \infty$. Indeed $b(\tau^2)$ grows logarithmically at infinity, whereas the pole and finite translation terms are bounded forms on $L^2(I_L)$. This explains the logarithmic form domain and semiboundedness of the localized form. For the logical proof target in Theorem S1.1, the smooth compactly supported class is enough.

S1.3 Shifted transfers and a sufficient contraction route

Define the entire completed zeta function and its centered version by

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad H(p) = \xi(\tfrac{1}{2} + p). \quad (\text{S1.10})$$

For $0 < \omega \leq 1/2$, the shifted transfer is

$$K_\omega(p) = \frac{H(p - \omega)}{H(p + \omega)}. \quad (\text{S1.11})$$

The functional equation and reality imply $|K_\omega(i\tau)| = 1$ where the ratio is defined. This boundary identity is unconditional; it is not a proof that an unweighted convolution operator is a contraction. In particular poles away from that boundary cannot be ignored.

The causal interpretation uses inverse Laplace transform on a line $\text{Re } p = \sigma > 1$. Let k_ω be that causal distribution and $V_{\omega,L} = P_L(k_\omega *) P_L$, where P_L denotes restriction to I_L together with zero extension when used as an input. The explicit kernel below is locally integrable for $\omega > 0$, so these finite-window operators are bounded. At $\omega = 0$, $k_0 = \delta_0$ and $V_{0,L} = I$. Logarithmic differentiation gives

$$\partial_\omega K_\omega(p) = -a_\omega(p) K_\omega(p), \quad a_\omega(p) = \frac{H'}{H}(p - \omega) + \frac{H'}{H}(p + \omega). \quad (\text{S1.12})$$

At zero shift the causal generator has Laplace symbol

$$a_0(p) = \frac{2}{p + 1/2} + \frac{2}{p - 1/2} - \log \pi + \psi(\tfrac{1}{4} + \tfrac{p}{2}) - 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-p \log n}. \quad (\text{S1.13})$$

The identity

$$\psi(\tfrac{1}{4} + \tfrac{p}{2}) - \psi(\tfrac{1}{4}) = 2 \int_0^\infty n_\Gamma(u) (1 - e^{-pu}) du \quad (\text{S1.14})$$

shows that the real quadratic form of this causal generator is exactly (S1.4). The two rational terms give the symmetric kernel $2 \cosh((x - y)/2)$; its form is $2|C|^2 - 2|S|^2$. On smooth test functions the first variation is therefore

$$\langle f, (I - V_{\omega,L}^* V_{\omega,L}) f \rangle = 2\omega Q_{0,L}[f] + o(\omega), \quad \omega \downarrow 0. \quad (\text{S1.15})$$

One obtains this expansion by differentiating the causal kernel at $\omega = 0$ using (S1.13); smooth zero extensions make the subtracted difference integral convergent in L^2 on a finite window. Thus an independent proof of $\|V_{\omega,L}\| \leq 1$ for every L and shifts tending to zero would imply RH. We need only this sufficient implication here; no global unweighted realization of K_ω is assumed.

S1.4 The exact positive part, and what it leaves unresolved

Write $a = 1/2 - \omega$, $b = 1/2 + \omega$, $s = 1/2 + p$, and $\mathcal{L}(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$, distinguished from the von Mangoldt function. Direct factorization yields

$$K_\omega(p) = \frac{(p+a)(p-b)}{(p+b)(p-a)} \widetilde{K}_\omega(p), \quad \widetilde{K}_\omega(p) = \frac{\mathcal{L}(s-\omega)}{\mathcal{L}(s+\omega)}. \quad (\text{S1.16})$$

The beta integral and Euler product, for $\text{Re } p > b$, give

$$g_\omega(t) = \frac{2\pi^\omega}{\Gamma(\omega)} e^{-at} (1 - e^{-2t})^{\omega-1} \mathbf{1}_{t>0}, \quad (\text{S1.17})$$

$$c_n(\omega) = n^{\omega-1/2} \prod_{\ell|n, \ell \text{ prime}} (1 - \ell^{-2\omega}), \quad c_1 = 1, \quad (\text{S1.18})$$

$$\widetilde{k}_\omega = g_\omega * \sum_{n \geq 1} c_n(\omega) \delta_{\log n} \geq 0. \quad (\text{S1.19})$$

For example, the substitution $r = e^{-2t}$ in the Laplace transform of (S1.17) gives $\pi^\omega \Gamma((p+a)/2) / \Gamma((p+b)/2)$. For the zeta quotient the coefficient at a prime power ℓ^k is $\ell^{k\omega} (1 - \ell^{-2\omega})$ before multiplication by $n^{-1/2}$, proving (S1.18) by multiplicativity. Only finitely many shifts contribute on a fixed window. These formulas define positive locally finite measures, not normalized probability laws; the endpoint $a = 0$ in particular precludes an unqualified Beta($a/2, \omega$) probability interpretation.

Absorb one rational factor into the positive part:

$$\begin{aligned} \widehat{k}_\omega &= \widetilde{k}_\omega + 2a(e^a \cdot \mathbf{1}_{>0}) * \widetilde{k}_\omega \geq 0, \\ k_\omega &= \widehat{k}_\omega - 2b(e^{-b} \cdot \mathbf{1}_{>0}) * \widehat{k}_\omega. \end{aligned} \quad (\text{S1.20})$$

Thus the full kernel is a positive part minus an exponential average. This exact reduction was central to the parent Loewner investigation [14]. It neither makes k_ω positive nor bounds the norm of its finite-window convolution. The required cancellation is still the arithmetic problem.

The fractional-dimension interpretation follows from the radial beta integral

$$\int_{\mathbb{R}^d} (1 + |u|^2)^{-(p+b)/2} d^d u = \pi^{d/2} \frac{\Gamma((p+b-d)/2)}{\Gamma((p+b)/2)}. \quad (\text{S1.21})$$

For integer d this is an ordinary integral in its convergence region; its radial continuation at $d = 2\omega$ reproduces the gamma factor. It is not a continuously variable lattice dimension. To see a concrete obstruction to one natural lattice continuation, put $\vartheta(q) = \sum_{k \in \mathbb{Z}} q^{k^2}$ and $m = 1 + d$. The coefficient of q^3 in $\vartheta(q)^m$ is

$$8 \binom{m}{3} = \frac{4}{3} m(m-1)(m-2) = \frac{4}{3} (d^3 - d) < 0 \quad (0 < d < 1). \quad (\text{S1.22})$$

It cannot be a positive point count. This excludes that interpolation, not every positive kernel or analytic continuation. Neither (S1.21) nor a bare Loewner map supplies the prime comb and rational subtraction in (S1.19)–(S1.20).

S2 Why defect matter reproduces the archimedean exponents

The geometric reason for using three-dimensional defect matter can be seen without supersymmetry. Normalize a free massless scalar covariance on the defect to $|X - Y|^{-1}$; coupling and color constants will be restored when needed. Write $X = e^x \Omega$, $Y = e^y \Omega'$ with $\Omega, \Omega' \in S^2$, and rescale the field by $e^{x/2}$. With $u = x - y$ and $v = \Omega \cdot \Omega'$, the covariance becomes

$$C(u, v) = \frac{1}{\sqrt{2(\cosh u - v)}} = \sum_{\ell=0}^{\infty} e^{-(\ell+1/2)|u|} P_{\ell}(v), \quad u \neq 0. \quad (\text{S2.1})$$

This is the Legendre generating function with parameter $e^{-|u|}$.

Fix an axis $e \in S^2$ and $0 \leq \rho < 1$. The Poisson density on S^2 and its even part are

$$P_{\rho}(v) = \frac{1 - \rho^2}{4\pi(1 - 2\rho v + \rho^2)^{3/2}}, \quad h_{\rho}(\Omega) = \frac{1}{2}\{P_{\rho}(e \cdot \Omega) + P_{\rho}(-e \cdot \Omega)\}. \quad (\text{S2.2})$$

Integrating (S2.1) against $h_{\rho}(\Omega)h_{\rho}(\Omega')$ projects onto even angular degrees. Orthogonality of the Legendre polynomials gives

$$C_{\rho}(u) = \sum_{n=0}^{\infty} \rho^{4n} e^{-a_n|u|} = \frac{e^{-|u|/2}}{1 - \rho^4 e^{-2|u|}}, \quad a_n = 2n + \frac{1}{2}. \quad (\text{S2.3})$$

For $\rho < 1$ this is a bounded continuous positive-definite covariance. For $u \neq 0$ it tends to $n_{\Gamma}(|u|)$ as $\rho \uparrow 1$. The limit is singular at $u = 0$, so positivity of the regulated covariance cannot simply be passed to a finite contact-subtracted covariance.

Proposition S2.1 (The sign of the subtraction). *Let T_{ρ} be convolution by C_{ρ} on $\mathbb{R}$, and put $M_{\rho} = \widehat{C}_{\rho}(0)$. Then $M_{\rho}I - T_{\rho}$ is positive, and its quadratic form tends to $\mathcal{E}_{\Gamma}$ on smooth compactly supported functions. The opposite subtraction $T_{\rho} - M_{\rho}I$ tends to the negative of that form. Adding any finite constant multiple of the identity to the latter limit does not give a positive form on all smooth compactly supported functions.*

Proof. Fourier transformation gives

$$\begin{aligned} \widehat{C}_{\rho}(\tau) &= \sum_{n \geq 0} \rho^{4n} \frac{2a_n}{a_n^2 + \tau^2}, \\ M_{\rho} - \widehat{C}_{\rho}(\tau) &= 2 \sum_{n \geq 0} \rho^{4n} \frac{\tau^2}{a_n(a_n^2 + \tau^2)} \uparrow b(\tau^2). \end{aligned} \quad (\text{S2.4})$$

Monotone convergence and (S1.7) prove the first claims. Moreover $M_{\rho} = \sum_{n \geq 0} \rho^{4n} / (n + 1/4)$ diverges logarithmically. Since $b(\tau^2) \rightarrow \infty$, any multiplier $c - b(\tau^2)$ is negative at high frequencies. Modulating a fixed nonzero compactly supported smooth function to arbitrarily high frequency gives a negative form value, also on any fixed nonempty interval. $\square$

There is a second useful decomposition. For $u \neq 0$ define the same-ray and opposite-ray functions

$$G_s(u) = \frac{1}{2 \sinh(|u|/2)}, \quad G_o(u) = \frac{1}{2 \cosh(u/2)}. \quad (\text{S2.5})$$

Then $n_{\Gamma}(|u|) = \frac{1}{2}(G_s(u) + G_o(u))$. The reflected free network constructed in Section S7 gives G_o . That is a positive and finite kernel, but it is not the even tower by itself. Conversely, the full

angular smearing above is an ordinary covariance construction with support on all of S^2 . It is not automatically a state supported in a positive-time half-space, as required for a time-reflection argument. Restricting the angular profile to a hemisphere changes its spectral weights.

The useful conclusion is therefore precise: free defect matter supplies the desired exponents and a regulated positive covariance. The RH target requires the associated difference energy, its fixed local term, its pole term and its arithmetic translations. These are distinct operations on the free data.

S2.1 An auxiliary Gaussian realization of the gamma ratio

The parent endpoint work also gave a useful fixed-space alternative to fractional dimension [13]. It can be understood from elementary Gaussian integration. Retain one complex mode for each $a_n = 2n + 1/2$, and for real p with $a_0 + p > \omega$ define

$$Z_N(p) = \prod_{n=0}^{N-1} \int_{\mathbb{C}} \frac{d^2 z_n}{\pi} e^{-(a_n+p)|z_n|^2} = \prod_{n=0}^{N-1} (a_n + p)^{-1}. \quad (\text{S2.6})$$

For a half-line field $\phi_n(t)$ with action $\int_0^\infty (|\phi'_n|^2 + a_n^2 |\phi_n|^2) dt$ and boundary value z_n , the decaying classical solution is $\phi_n(t) = z_n e^{-a_n t}$. Its action is $a_n |z_n|^2$. Adding the boundary term $p|z_n|^2$ gives the same Gaussian; varying the boundary value gives the Robin condition $(-\partial_t + p)\phi_n(0) = 0$ for the full variational problem. Constants common to the two partition functions cancel in the ratio.

Set $x = (p + 1/2 - \omega)/2$ and $y = x + \omega$. The finite product gives

$$\frac{Z_N(p - \omega)}{Z_N(p + \omega)} = \prod_{n=0}^{N-1} \frac{n + y}{n + x} = \frac{\Gamma(N + y)\Gamma(x)}{\Gamma(y)\Gamma(N + x)}. \quad (\text{S2.7})$$

Since $\Gamma(N + y)/\Gamma(N + x) \sim N^\omega$,

$$\pi^\omega \lim_{N \rightarrow \infty} N^{-\omega} \frac{Z_N(p - \omega)}{Z_N(p + \omega)} = \pi^\omega \frac{\Gamma((p + a)/2)}{\Gamma((p + b)/2)}. \quad (\text{S2.8})$$

Equivalently one can divide by the same finite product at a reference p_0 to cancel the normalization. One real Gaussian mode would give the square root of the ratio, so statistics and multiplicities matter.

This is an auxiliary partition-function model, not a localization of the Wilson network. Angular smearing samples a channel; it does not remove the other angular modes from a partition function. The full scalar sphere has multiplicity $2\ell + 1$, whereas the model retains one copy of each even axial mode. A physical projection or cancellation of the extra modes has not been derived. Here p is a boundary mass, and identifying it with the arithmetic Laplace frequency requires a dynamical dictionary. The Robin response $(a_n + p)^{-1}$, the partition ratio, and a causal transfer are different objects. The supplied factor π^ω , the prime comb, and the rational pole correction remain outside this Gaussian mechanism.

S3 Semicircles, constant drivers, and a restricted rigidity theorem

Choose a plane containing a direction tangent to the defect and its normal, and write $z = x + iy$ with $y > 0$. The common-reference semicircle from $r > 0$ to zero is

$$z_r(\theta) = \frac{r}{2}(1 + \cos \theta + i \sin \theta), \quad 0 \leq \theta < \pi. \quad (\text{S3.1})$$

Inversion preserves the upper half-plane and gives

$$w_r(\theta) = -\frac{1}{z_r(\theta)} = -\frac{1}{r} + \frac{i}{r} \tan(\theta/2). \quad (\text{S3.2})$$

This is a vertical slit. In the chordal convention

$$\partial_t g_t(w) = \frac{2}{g_t(w) - U_t}, \quad g_0(w) = w, \quad (\text{S3.3})$$

the constant driver $U_t = a = -1/r$ has solution

$$g_t(w) = a + \sqrt{(w - a)^2 + 4t}, \quad \gamma_t = a + 2i\sqrt{t}, \quad z_t = -\frac{1}{a + 2i\sqrt{t}}. \quad (\text{S3.4})$$

The square-root branch is fixed by $g_t(w) \sim w$ at infinity. Its expansion is $g_t(w) = w + 2t/w + O(w^{-2})$, so the half-plane capacity is $2t$, consistent with [7, Lecture 2, Proposition 2.17]. Comparing the traces gives the exact dictionary

$$t = \frac{\tan^2(\theta/2)}{4r^2}. \quad (\text{S3.5})$$

The complete semicircle reaches the reference only as $t \rightarrow \infty$. At fixed angle, $r \mapsto \lambda r$ sends $t \mapsto \lambda^{-2}t$. Thus this inverted capacity is neither the coordinate $\log r$ nor the arithmetic shift ω . The formulas identify contours; they do not assert equality of renormalized expectations through an inversion singular at an endpoint.

S3.1 Fixed scalar coupling and fixed special charge

Let the Euclidean Clifford generators satisfy $\{\Gamma_A, \Gamma_B\} = 2\delta_{AB}$. Take one internal direction I and fix a nonzero complex spinor ϵ_c obeying

$$\epsilon_s = 0, \quad J\epsilon_c = -\epsilon_c, \quad J = i\Gamma_I\Gamma_3. \quad (\text{S3.6})$$

Here $\epsilon_s + X^\mu \Gamma_\mu \epsilon_c$ is the relevant conformal Killing spinor, with its constant part set to zero. These are the inherited common bulk conditions. For a regular real curve $X = (x, y) \neq 0$ in the $(1, 3)$ -plane and constant positive scalar arclength coupling, its local line condition is

$$(i\dot{x}\Gamma_1 + i\dot{y}\Gamma_3 + v\Gamma_I)(x\Gamma_1 + y\Gamma_3)\epsilon_c = 0, \quad v = |\dot{X}| > 0. \quad (\text{S3.7})$$

Proposition S3.1 (Rigidity under (S3.6)). *Equation (S3.7) holds if and only if*

$$\frac{\dot{z}}{|\dot{z}|} = \frac{iz^2}{|z|^2}. \quad (\text{S3.8})$$

Consequently $-1/z$ has constant real part and increasing imaginary part. Connected solutions in the upper half-plane are subarcs of circles through zero with center on the real axis, or the degenerate imaginary-axis line. A solution from a nonzero real boundary point to zero is the corresponding oriented upper semicircle.

Proof. The projector implies $\Gamma_I \epsilon_c = -i\Gamma_3 \epsilon_c$. Clifford multiplication reduces (S3.7) to $(A + B\Gamma_1\Gamma_3)\epsilon_c = 0$, where

$$A = x\dot{x} + y\dot{y} + vy, \quad B = y\dot{x} - x\dot{y} + vx. \quad (\text{S3.9})$$

Since $(\Gamma_1\Gamma_3)^2 = -1$, multiplication by $A - B\Gamma_1\Gamma_3$ gives $(A^2 + B^2)\epsilon_c = 0$. The coefficients are real and the spinor is nonzero, so $A = B = 0$. Solving gives

$$\frac{\dot{x}}{v} = -\frac{2xy}{x^2 + y^2}, \quad \frac{\dot{y}}{v} = \frac{x^2 - y^2}{x^2 + y^2}, \quad (\text{S3.10})$$

which is (S3.8). Conversely it makes $A = B = 0$. Moreover $d(-1/z)/ds = iv/|z|^2$. The level set $\text{Re}(-1/z) = a \neq 0$ is $x^2 + y^2 = -x/a$, and $a = 0$ gives the imaginary axis. $\square$

For instance, an inverted deformation $w(s) = a + \varepsilon h(s) + is$ preserves these conditions only if $\varepsilon h'(s) = 0$. A constant h changes the slit location; a nontrivial fixed-endpoint deformation fails. A regular capacity-parametrized Loewner trace satisfying the same conditions therefore has constant driving.

This is a bulk theorem for one plane, one charge and one scalar prescription. It is not a classification of supersymmetric contours with varying scalar directions, other conformal Killing spinors, additional endpoint degrees of freedom, or nonplanar embeddings. Section S5 supplies a constructive control showing that a changed scalar prescription allows arbitrary tangents.

S4 Smooth Wilson variation and the Loewner insertion

S4.1 Conventions and the exact variation

Work with smooth finite matrix fields and a regular C^2 curve $X : [0, 1] \rightarrow \mathbb{R}^4$. Let A_μ and the scalar fields X_I be Hermitian, and let $n^I(s)$ be a unit internal vector. Define

$$\mathcal{M}(s) = iA_\mu(X(s))\dot{X}^\mu(s) + \Phi(X(s), s)v(s), \quad \Phi = n^I X_I, \quad v = |\dot{X}|. \quad (\text{S4.1})$$

The ordered transport $U(b, a)$ solves $\partial_b U(b, a) = \mathcal{M}(b)U(b, a)$, $U(a, a) = I$; larger parameters stand to the left. Our curvature and covariant derivative are

$$F_{\nu\mu} = \partial_\nu A_\mu - \partial_\mu A_\nu - i[A_\nu, A_\mu], \quad D_\nu \Phi = \partial_\nu \Phi - i[A_\nu, \Phi]. \quad (\text{S4.2})$$

Vary the curve by $\eta = \delta X$ and the internal vector by δn , holding the spacetime fields fixed.

Proposition S4.1 (Ordered transport variation). *With $C_s = iA_\nu(X(s))\eta^\nu(s)$,*

$$\delta U(1, 0) = C_1 U(1, 0) - U(1, 0)C_0 + \int_0^1 U(1, s)\mathcal{I}(s)U(s, 0)ds, \quad (\text{S4.3})$$

$$\mathcal{I} = iF_{\nu\mu}\eta^\nu \dot{X}^\mu + vD_\nu \Phi \eta^\nu + \Phi \frac{\dot{X} \cdot \dot{\eta}}{v} + vX_I \delta n^I. \quad (\text{S4.4})$$

Proof. Differentiating the defining ODE gives $\delta U = \int U \delta \mathcal{M} U$. Direct differentiation of $\mathcal{M}$ yields $\delta \mathcal{M} = \dot{C} + [C, \mathcal{M}] + \mathcal{I}$. Meanwhile

$$\frac{d}{ds} \{U(1, s)C_s U(s, 0)\} = U(1, s)(\dot{C} + [C, \mathcal{M}])U(s, 0). \quad (\text{S4.5})$$

Integration proves the formula, including its ordering and boundary signs. $\square$

For constant n and fixed endpoints, integrate the arclength term by parts. Writing $T = dX/d\ell$, $k = dT/d\ell$, one obtains

$$\begin{aligned}\delta U &= \int_C U(1, s) \eta^\nu \mathcal{D}_\nu U(s, 0) d\ell, \\ \mathcal{D}_\nu &= iF_{\nu\mu} T^\mu + (\delta_\nu^\mu - T_\nu T^\mu) D_\mu \Phi - \Phi k_\nu.\end{aligned}\tag{S4.6}$$

In particular $T^\nu \mathcal{D}_\nu = 0$, as required for a fixed-endpoint reparametrization. The curvature term is necessary because the scalar couples to arclength. If n varies, one must retain its additional terms in (S4.4).

Let the endpoints be $x = X(0)$ and $y = X(1)$, both on the defect, and consider $\mathcal{O} = \bar{q}(y)U(1, 0)q(x)$. The boundary terms in (S4.3) combine with endpoint differentiation to give

$$\begin{aligned}\delta \mathcal{O} &= \eta_y^\nu (\partial_\nu \bar{q} + i\bar{q}A_\nu)(y)Uq(x) + \bar{q}(y)U\eta_x^\nu (D_\nu q)(x) \\ &\quad + \int_0^1 \bar{q}(y)U(1, s)\mathcal{I}(s)U(s, 0)q(x) ds.\end{aligned}\tag{S4.7}$$

Here $D_\nu q = \partial_\nu q - iA_\nu q$, and endpoint displacements are tangent to the defect. Changing endpoint R polarizations contributes further terms. A defect field q is not defined at a bulk tip.

S4.2 Regular inverse-map deformations

Let $f_t = g_t^{-1}$ for (S3.3), and choose a fixed smooth reference contour $\zeta(s)$ in a region where f_t is regular and $\zeta(s) \neq U_t$. Differentiating $g_t(f_t(w)) = w$ gives

$$X_t(s) = f_t(\zeta(s)), \quad \eta_t(s) = \partial_t X_t(s) = -\frac{2f_t'(\zeta(s))}{\zeta(s) - U_t}.\tag{S4.8}$$

Insert its real and imaginary components into (S4.4) and (S4.7). This is a concrete field-dependent evolution of the transport on a pulled-back test curve. It is not a variation formula for the singular growing tip. Regular real boundary segments remain on the defect in this planar interpretation. A growing prefix instead carries an uncontracted color index at its interior endpoint; it requires a specified tip state or a conditional completion of the curve.

If $dU_t = \sqrt{\kappa} dB_t$ for standard Brownian motion B_t , then at fixed w the inverse-map equation is of finite variation until a singularity is reached. In centered coordinates $\mathfrak{f}_t(w) = f_t(w + U_t)$, stopped smooth Ito calculus gives

$$d\mathfrak{f}_t(w) = \left[-\frac{2\mathfrak{f}_t'(w)}{w} + \frac{\kappa}{2}\mathfrak{f}_t''(w) \right] dt + \sqrt{\kappa}\mathfrak{f}_t'(w) dB_t.\tag{S4.9}$$

This is only a map identity. No scalar arclength integral or quantum Wilson transport on an unregulated stochastic trace is defined by it.

S4.3 A precise obstruction to geometry-only closure

Proposition S4.2. *There is no universal multiplicative shape generator for a single Wilson transport whose coefficient depends only on the curve and its deformation, independently of the background fields.*

Proof. In an abelian theory take $A_x = cy$, $A_y = \Phi = 0$ and the straight path $X(s) = (s, 0)$, $0 \leq s \leq 1$. Then $U = 1$ for every real c . For $\eta = (0, h(s))$ with $h(0) = h(1) = 0$,

$$\delta U = ic \int_0^1 h(s) ds. \quad (\text{S4.10})$$

The same curve, deformation and original transport thus have different derivatives. As a positive control, a pure gauge $A = d\chi$ has fixed-endpoint transport independent of the contour. $\square$

The proposition does not exclude closure of an averaged correlation function after field equations, Ward identities or a protected reduction are supplied. It identifies the extra information such a closure must use. At leading free endpoint order $U = I$, fixed-endpoint interior deformations do not change (S2.3); the nontrivial information must enter at higher order or through another specified sector.

S4.4 The tangent connection as a one-form

For a constant real isometry $M : \mathbb{R}^4 \rightarrow \mathbb{R}^6$, set $n = MT$, so that $\Phi|dX| = \Phi_\mu dX^\mu$ with $\Phi_\mu = M^A{}_\mu X_A$. Define $\mathcal{B}_\mu = iA_\mu + \Phi_\mu$. Then the fixed-endpoint variation takes the ordinary one-form form

$$\begin{aligned} \delta U_M &= \int U_M(1, s) \mathcal{F}_{\nu\mu} \eta^\nu \dot{X}^\mu U_M(s, 0) ds, \\ \mathcal{F}_{\nu\mu} &= \partial_\nu \mathcal{B}_\mu - \partial_\mu \mathcal{B}_\nu - [\mathcal{B}_\nu, \mathcal{B}_\mu] \\ &= iF_{\nu\mu} + D_\nu \Phi_\mu - D_\mu \Phi_\nu - [\Phi_\nu, \Phi_\mu]. \end{aligned} \quad (\text{S4.11})$$

For moving endpoints the boundary insertion is $\mathcal{B}_\nu \eta^\nu$. These signs follow from the same ordered ODE. Supersymmetry of the line is not an assertion that this complex curvature vanishes on arbitrary field configurations.

S5 Tangent coupling with the full chiral defect projector

S5.1 Field content and conventions

The defect is $x^3 = 0$ in Euclidean spacetime (x^0, x^1, x^2, x^3) . The bulk has a gauge field A_μ , six adjoint scalar fields X_A , $A = 4, \dots, 9$, and their supersymmetric fermion partners. The triplets (X_4, X_5, X_6) and (X_7, X_8, X_9) transform under $SU(2)_H$ and $SU(2)_V$, respectively. Defect matter includes fundamental scalars q_m , $m = 1, 2$, forming an H doublet, and fermions forming a V doublet; $\bar{q}_m$ are the physical antifundamental conjugates. These are the bulk/defect assignments of [4, 6].

Work in a 32-dimensional complex Euclidean Clifford module, with

$$\chi = -i\Gamma_0\Gamma_1 \cdots \Gamma_9, \quad D = \Gamma_3\Gamma_4\Gamma_5\Gamma_6, \quad \chi\epsilon = \epsilon, \quad D\epsilon = \epsilon. \quad (\text{S5.1})$$

The second equation is the unfluxed D3–D5 intersection condition. It is the projector appearing in [5, eqs. (2.7), (4.1)–(4.3), (4.9)]; the terminating-brane boundary conditions and Nahm pole are not part of our assumptions. The displayed conventions fix the chirality and orientation signs. Complex ranks below are not counts of real Euclidean Majorana–Weyl supercharges.

S5.2 A normal plane

For the (x^1, x^3) -plane take the connection

$$\mathcal{B} = iA_1 dx^1 + iA_3 dx^3 + X_8 dx^1 + X_6 dx^3. \quad (\text{S5.2})$$

This is the tangent prescription $n = (\dot{x}^1 e_8 + \dot{x}^3 e_6)/|\dot{x}|$, so the scalar direction changes with orientation. In the usual local supersymmetry variation of scalar-coupled transport, a constant Poincare parameter is annihilated by every tangent if

$$[\dot{x}^1(i\Gamma_1 + \Gamma_8) + \dot{x}^3(i\Gamma_3 + \Gamma_6)]\epsilon = 0. \quad (\text{S5.3})$$

This is the tangent construction of [3], now with (S5.1) imposed as well. Define

$$K_1 = i\Gamma_8\Gamma_1, \quad K_n = i\Gamma_6\Gamma_3. \quad (\text{S5.4})$$

Proposition S5.1 (Planar charge space). *The common subspace $\chi = D = +1$, $K_1 = K_n = -1$ has complex dimension two. Every parameter in it obeys (S5.3) for every smooth planar tangent.*

Proof. The four displayed matrices are commuting Hermitian involutions. Their orthogonal projector is

$$P_{\text{plane}} = \frac{1}{16}(1 + \chi)(1 + D)(1 - K_1)(1 - K_n). \quad (\text{S5.5})$$

Every nonidentity Clifford word in its expansion has zero trace, and no nonempty product of these four generators is scalar. Hence $\text{tr } P_{\text{plane}} = 32/16 = 2$. Multiplication of $K_1\epsilon = -\epsilon$ by Γ_8 , and of $K_n\epsilon = -\epsilon$ by Γ_6 , gives the two linear equations in (S5.3) separately. $\square$

The triplet assignments are necessary within the stated real, constant-map ansatz. The defect projector D commutes with a defect tangent gamma and with a V gamma; it anticommutes with the normal gamma and with an H gamma. Pairing a tangent with H, or the normal with V, therefore gives an involution anticommuting with D and no simultaneous eigenspinor. More generally, split a proposed real scalar vector into H and V components. For an independently variable defect tangent its wrong H component lies in the opposite D -eigenspace from the remaining line equation, so it must annihilate ϵ separately. Its Clifford square is its squared real length, making it invertible unless that component is zero. The normal argument exchanges H and V. Complex null internal vectors and couplings restricted to one constrained tangent are outside this argument. Proposition S5.1 is the nonzero control.

S5.3 One charge across all defect directions

To accommodate an angular family, introduce three tangent conditions

$$K_\mu^{(s)} = is_\mu \Gamma_{7+\mu} \Gamma_\mu, \quad \mu = 0, 1, 2, \quad s_\mu \in \{+1, -1\}. \quad (\text{S5.6})$$

The corresponding oriented tangent isometry has determinant $s_0 s_1 s_2$. Direct Clifford multiplication gives

$$K_0^{(s)} K_1^{(s)} K_2^{(s)} = (s_0 s_1 s_2) \chi D. \quad (\text{S5.7})$$

Proposition S5.2 (Orientation and the all-direction charge). *For chirality $\chi = c$ and defect sign $D = d$, all three tangent eigenvalues can be -1 , together with $K_n = -1$, if and only if the tangent isometry has determinant $-cd$. With that sign the common space has complex dimension one; with the opposite sign it is zero.*

Proof. Taking eigenvalues in (S5.7) proves necessity. For the allowed sign, one tangent condition follows from the other two and χ, D . The remaining five commuting conditions $\chi = c$, $D = d$, $K_n = -1$, $K_0^{(s)} = K_1^{(s)} = -1$ are independent. Their projector has trace $32/2^5 = 1$, by the same nonidentity-word trace argument as above. Orthogonal changes of the oriented tangent and internal frames give the general isometry statement. $\square$

For $c = d = +1$, choose once and for all

$$Me_0 = -e_7, \quad Me_1 = e_8, \quad Me_2 = e_9, \quad Me_3 = e_6. \quad (\text{S5.8})$$

Then $M^T M = I$ and the connection is

$$\mathcal{B} = iA_\mu dx^\mu - X_7 dx^0 + X_8 dx^1 + X_9 dx^2 + X_6 dx^3, \quad \mu = 0, 1, 2, 3. \quad (\text{S5.9})$$

The same nonzero constant complex parameter works for every Euclidean four-dimensional tangent, hence for all normal planes indexed by S^2 . This is not a statement about arbitrary real Lorentzian trajectories with real internal couplings.

S5.4 The remaining H weight

Set $T_H = i\Gamma_4\Gamma_5$. The exact identity

$$D = K_n T_H \quad (\text{S5.10})$$

implies $T_H \epsilon = -\epsilon$ for our signs. The tangent projectors reduce the spacetime/V factors but do not introduce the other H weight. Thus a surviving constant charge has the same one-dimensional H factor at both endpoints. Using Baker's generator convention, the plus-normal combination $iA_3 + X_6$ is labeled by preserved generators Q_{i2} ; the conjugate convention exchanges 1 and 2. Spinor-parameter and generator indices are related by charge conjugation, so the endpoint labels must be fixed from the physical transformations, not read off from a gamma eigenvalue alone.

S6 Endpoint polarization, vanishing, and the adjoint

S6.1 The same-charge endpoint calculation

The physical scalar variations may be written $\delta q_m = 2\bar{\zeta}_{im}\Psi_i$ and $\delta \bar{q}_m = 2\bar{\Psi}_i\zeta_{im}$, with the related physical generator conjugation; see [6, Appendix C]. Complexification does not make the actions on q and $\bar{q}$ independently adjustable. For the plus-normal choice the annihilated scalar components are q_2 and $\bar{q}_1$.

Here is a basis-independent polarization version of that statement. Suppress the nonzero spacetime/V factor of the charge, let its H factor be $a \in \mathbb{C}^2 \setminus \{0\}$, and write

$$q(v) = \sum_m v_m q_m, \quad \bar{q}(w) = \sum_m w_m \bar{q}_m, \quad E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (\text{S6.1})$$

In the generator convention just specified, the invariant tensors give

$$Q(a)q(v) \propto (a^T E v)\Psi, \quad Q(a)\bar{q}(w) \propto (a^T w)\tilde{\Psi}. \quad (\text{S6.2})$$

This follows by rotating the representative conditions $Q_{i2}q_2 = Q_{i2}\bar{q}_1 = 0$, with E the fundamental antisymmetric tensor and the second contraction the fundamental/dual evaluation. The suppressed fermion components are arbitrary, so an operator annihilator requires the scalar coefficient to vanish.

Proposition S6.1 (Zero same-charge scalar pairing). *For the constant fixed-H-factor charge above, nonzero ordinary fundamental and antifundamental scalar endpoints separately killed by that charge have zero mutual free contraction. They are not a physical adjoint pair.*

Proof. The two nullspaces in (S6.2) are

$$v = \lambda a, \quad w = \mu E a. \quad (\text{S6.3})$$

The H tensor of the free propagator is δ_{mn} , so its polarization factor is $v^T w = \lambda \mu a^T E a = 0$. Physical conjugation instead requires $w = v^*$ and gives $v^T v^* = \|v\|^2 > 0$. If $v = \lambda a \neq 0$, then $a^T v^* = \lambda^* \|a\|^2 \neq 0$, so the physical conjugate is not killed by that same charge. $\square$

In the representative basis, q_2 pairs nontrivially with $\bar{q}_2$, whereas its same-charge partner is $\bar{q}_1$. The physical adjoint pair is a useful nonzero control, but its dressed expectation still requires an independent positivity argument. At distinct endpoints the two supersymmetry variations contain independent fermion fields, so they cannot cancel each other as bare operator identities. In this ansatz the bulk line variation already vanishes separately. Junction matter or a fermionic superconnection would change the operator and require a new calculation.

S6.2 The full bilinear has nonzero R charge

The fields in (S5.2) and (S5.9) are neutral under the H rotation fixing X_6 . Assign q_1, q_2 weights $+1/2, -1/2$. Then $\bar{q}_1$ also has weight $-1/2$ and

$$\mathcal{O}_C = \bar{q}_1(b) U_M(C) q_2(a) \quad (\text{S6.4})$$

has charge -1 . If the vacuum and the definition of the operator preserve this symmetry, its expectation obeys

$$\langle \mathcal{O}_C \rangle = e^{-i\alpha} \langle \mathcal{O}_C \rangle \quad \text{for every } \alpha, \quad \langle \mathcal{O}_C \rangle = 0. \quad (\text{S6.5})$$

This selection rule extends beyond the free contraction. It does not construct an ultraviolet-finite line operator. A compensating charged insertion, a different endpoint sector or a symmetry-breaking vacuum changes its assumptions. A fixed-endpoint deformation of the same neutral connection carries no compensating charge.

S6.3 Why the original semicircle is a nonzero control

The restriction is to a constant Poincare charge and the displayed tangent map. On (S3.1), that map has scalar direction $n_8 = -\sin \theta$, $n_6 = \cos \theta$: the scalar normal direction changes sign along with the tangent. The common Poincare charge therefore keeps its H weight fixed.

The original semicircle with constant scalar arclength coupling instead uses a position-dependent conformal Killing spinor. Its local endpoint conditions can select different H factors at the two ends. Baker's same-polarization semicircle has a nonzero leading endpoint contraction, proportional to $g^2/(4\pi|a-b|)$ in that paper's normalization [6, Section 3.2]. It is consequently a successful control outside Proposition S6.1, rather than a counterexample to it. This distinction is also why the rigidity theorem and tangent-map construction must not be conflated.

S6.4 The adjoint changes the scalar map

For a tangent-coupled line and Hermitian matrix fields,

$$U_M(C)^\dagger = U_{-M}(C^{-1}), \quad U_M(C^{-1}) = U_M(C)^{-1}. \quad (\text{S6.6})$$

To prove the first identity, reverse the product order in the discretized ordered exponential and conjugate each exponent. The gauge term changes sign under conjugation; after reversing the contour it is again $iA_\mu dx^\mu$, while the scalar one-form has acquired $-M$. The second identity is ordinary inverse transport. Thus contour reversal with unchanged tangent map gives the inverse, not the Hermitian adjoint. A reflected norm must implement the former identity with the appropriate spacetime pullback. This is the purpose of the next section.

S7 Physical reflection and the reference color junction

S7.1 Ordinary time reflection

Reflect a time direction tangent to the defect:

$$\vartheta x = Rx, \quad R = \text{diag}(-1, 1, 1, 1), \quad b = 0 \in \{x^0 = 0\} \cap \{x^3 = 0\}. \quad (\text{S7.1})$$

The positive half-space is $x^0 > 0$. In the bosonic adjoint convention, A transforms as a one-form, all six Hermitian scalars are even under geometric time reflection, and conjugation exchanges q_m with $\bar{q}_m$. Fermionic reflection and an interacting regulator are additional requirements, not consequences of these bosonic rules.

Let $C : a \rightarrow b$ lie in the positive half-space except at b , with a on the defect. Use M from (S5.8). Pullback of the fields first gives $U_{MR}(\vartheta C)$; applying (S6.6) then gives

$$\Theta_0 U_M(C) = U_N(\vartheta C^{-1}), \quad N = -MR. \quad (\text{S7.2})$$

Explicitly,

$$Ne_0 = -e_7, \quad Ne_1 = -e_8, \quad Ne_2 = -e_9, \quad Ne_3 = -e_6. \quad (\text{S7.3})$$

The time component retains its scalar map and the spatial components reverse it. This ordering calculation agrees with the distinction between geometric reflection and scalar-coupled adjoint transport in [10, eqs. (4)–(7)].

S7.2 The actual paired operator and the conditional Gram statement

Define a covariant color vector at the reference by

$$F_i^\alpha = [U_M(C_i)]^\alpha{}_\beta q_2^\beta(a_i). \quad (\text{S7.4})$$

Its reflected row is $\bar{q}_{2,\gamma}(\vartheta a_i)[U_N(\vartheta C_i^{-1})]^\gamma{}_\alpha$. The proposed kernel is

$$K_{ij} = \left\langle \bar{q}_{2,\gamma}(\vartheta a_i)[U_N(\vartheta C_i^{-1})]^\gamma{}_\alpha [U_M(C_j)]^\alpha{}_\beta q_2^\beta(a_j) \right\rangle. \quad (\text{S7.5})$$

There is an identity color contraction at b , not another defect field there. Under a gauge transformation, $F_i \mapsto g(b)F_i$ and its reflected row transforms by $g(b)^{-1}$, so the paired observable is gauge invariant. The individual vector is not a gauge-invariant state without a specified reference charge sector.

Assumption S7.1. A regulator and reference color sector admit Osterwalder–Schrader positivity for these half-space functionals. The regulated endpoint and junction operators belong to that domain, and their limiting renormalization preserves the inequality whenever a limit is claimed.

Proposition S7.2 (Conditional positivity). *Under the regulated part of Assumption S7.1, the matrix in (S7.5) is positive semidefinite for every finite collection of admissible contours and endpoints.*

Proof. Apply the assumed OS inequality to $\sum_i c_i F_i$, including its reference color fiber, to obtain $\sum_{ij} \bar{c}_i K_{ij} c_j \geq 0$. For a concrete bosonic sufficient model, suppose the two half-integrals factor over boundary data B with positive measure ν . They give vector wavefunctions $\psi_i^\alpha(B)$ and

$$K_{ij} = \int \sum_{\alpha} \overline{\psi_i^\alpha(B)} \psi_j^\alpha(B) d\nu(B), \quad \nu \geq 0. \quad (\text{S7.6})$$

The claimed inequality is then the integral of a squared norm. Boundary gauge transformations act unitarily on the displayed fiber. $\square$

This conditional statement locates the missing physical input. A positive bosonic factorization is not asserted for the full fermionic $\mathcal{N} = 4$ defect measure. Gauge invariance of (S7.5) alone proves no reflection inequality. Gauss law, the reference charge sector, the regulator, and junction counterterms must all be addressed in an interacting realization.

S7.3 An unconditional free kernel

At leading free endpoint order $U = I$. Strip the positive color and coupling constants and write $a_i = (\tau_i, \mathbf{z}_i, 0)$ with $\tau_i > 0$, $\mathbf{z}_i \in \mathbb{R}^2$. Then

$$K^{(0)}(a_i, a_j) = \frac{1}{\sqrt{(\tau_i + \tau_j)^2 + |\mathbf{z}_i - \mathbf{z}_j|^2}}. \quad (\text{S7.7})$$

Its representation

$$K^{(0)}(a_i, a_j) = 4\pi \int_{\mathbb{R}^2} \frac{d^2 k}{(2\pi)^2} \frac{e^{-|k|(\tau_i + \tau_j)} e^{ik \cdot (\mathbf{z}_j - \mathbf{z}_i)}}{2|k|} \quad (\text{S7.8})$$

is an integral of products $\overline{\psi_i(k)} \psi_j(k)$ with positive measure. This proves free positivity independently of Assumption S7.1. On the time ray $a_r = (r, 0, 0, 0)$ it reduces to

$$K^{(0)}(r, s) = \frac{1}{r + s} = \int_0^\infty e^{-\lambda r} e^{-\lambda s} d\lambda, \quad \sqrt{rs} K^{(0)}(r, s) = G_o(\log r - \log s). \quad (\text{S7.9})$$

The diagonal is finite for $r > 0$. The kernel is Hankel in r, s , and translation invariant after the radial weights and logarithmic coordinates. It is the opposite-ray kernel of (S2.5), not the full even archimedean tower. The full-sphere profile in Section S2 cannot be used as the same half-space state without changing its support and recomputing its weights.

S7.4 The two branches preserve different constant charges

Let $\widetilde{K}_\mu = i\Gamma_{Me_\mu} \Gamma_\mu$ for $\mu = 0, 1, 2$, and $K_n = i\Gamma_6 \Gamma_3$. With $\chi = D = +1$, the two branches require the following eigenvalues:

Invololution	Ket with M	Reflected bra with N
$\widetilde{K}_0$	-1	-1
$\widetilde{K}_1, \widetilde{K}_2$	-1	$+1$
K_n	-1	$+1$

Each all-direction family has one complex compatible charge; in the (x^0, x^3) -plane each has two. Their common subspace is zero when time and normal tangents are independently required, because the normal conditions are opposite. The normal projectors also select opposite H weights: the ket uses q_2 and the physical bra uses $\bar{q}_2$ with the conjugate-weight charge. The identity tensor at b cannot compensate a nonzero variation on a branch.

As a bulk control, a straight time-directed contour needs only $\widetilde{K}_0 = -1$. That condition is common to the maps and gives a four-dimensional complex chiral defect solution space. It is a bulk control, not a statement that a constant charge annihilates the physical adjoint pair of endpoint scalars.

S7.5 Internal rotations do not automatically preserve positivity

The proper rotations by π about H axis 4 and V axis 7 are

$$S_H = \text{diag}(1, -1, -1), \quad S_V = \text{diag}(1, -1, -1). \quad (\text{S7.10})$$

For $S = S_H \oplus S_V$, one checks $SN = M$. Thus an internally rotated reflection aligns the two bulk scalar maps. But S_H acts on the endpoint doublet by $\pm i\sigma_1$, up to basis convention. It takes $\bar{q}_2$ to a phase times $\bar{q}_1$, restoring the zero diagonal contraction of Proposition S6.1.

On the full free endpoint doublet the inserted metric is not positive. Without a phase adjustment its eigenvalues are $\pm i$; with a phase making it Hermitian they are $+1, -1$. For instance

$$(1, -1)\sigma_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} / 2 = -1. \quad (\text{S7.11})$$

No overall phase makes both eigenvalues positive. The identity metric from ordinary adjoint pairing is the control. More generally, if a unitary S on a positive Hilbert space satisfies $\langle \psi, S\psi \rangle \geq 0$ for every ψ , then S is positive self-adjoint; unitarity forces its spectrum to be $\{1\}$ and $S = I$. Projection to one rotation eigenspace is a different operation and here mixes the fixed H endpoint weights.

This is a counterexample for the displayed doublet state space. It does not contradict the special large-rectangle rotated-reflection conjecture distinguished from ordinary reflection in [10, eqs. (11)–(16)].

S7.6 A free test of an odd-H normal reflection

A different proposal in the parent investigation reflected across the defect and assigned odd parity to the H scalars. For the canonical free scalar algebra, a real mode of frequency $\nu > 0$ at distance $a > 0$ from a reflection plane has ordinary norm $e^{-2\nu a}/(2\nu) > 0$; assigning it odd reflection parity changes the sign. The failure is also visible in a gauge-invariant observable. For a non-Abelian group with $\text{tr}(T_a T_b) = \delta_{ab}/2$, take the Hermitian Wick polynomial

$$\mathcal{O} = i \text{tr}(H[V_1, V_2]), \quad (\text{S7.12})$$

where H is one H scalar and V_1, V_2 are distinct V scalars. Writing $[T_b, T_c] = if_{bca}T_a$, its coefficient is $-f_{bca}/2$. Distinct species have only their matching free contractions, hence at separated reflected points

$$\langle \Theta_{\text{even}} \mathcal{O}(x) \mathcal{O}(x) \rangle = \frac{1}{4} \sum_{abc} f_{abc}^2 D_0(\vartheta x - x)^3 > 0, \quad (\text{S7.13})$$

with D_0 the positive scalar covariance. Odd H and even V parity give the negative of this value. This excludes that reflection on the unrestricted canonical free bulk observable algebra. It does not classify all interacting normal-reflection actions or restricted observable sectors. The even assignment supplies the positive control.

S8 A smooth reference join and a direct bulk response

S8.1 A contour family that meets the reflection plane smoothly

Let v be the incoming unit ket tangent at b . Its reflected reverse has outgoing tangent $-Rv$. A smooth join requires

$$v = -Rv, \quad (\text{S8.1})$$

so v is parallel to the time direction e_0 . The internal scalar directions then agree as well, since $N(-Rv) = Mv$. For a kernel involving many contours, every ket should have the same reference tangent. The old semicircles arrive along $-e_3$ and return along $+e_3$ under time reflection, producing a backtracking join instead.

Contour freedom provides a simple replacement. In the (x^0, x^3) -plane set

$$X_{T,\eta}(t) = (Tt, \eta T f(t)), \quad f(t) = t^2(1-t), \quad 0 \leq t \leq 1, \quad T > 0, \quad \eta \geq 0. \quad (\text{S8.2})$$

The ket traverses the curve from $t = 1$ to zero. The endpoints are on the defect, its interior is in positive time and, for $\eta > 0$, strictly above the defect. Since $f'(0) = 0$, its incoming unit tangent is $-e_0$. The reflected return is regular with a continuous scalar direction even when the two branches have different T or η . Curvature need not agree.

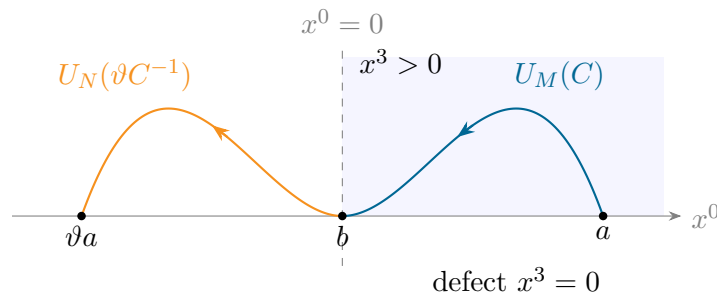


Figure S1: A polynomial bulge and its ordinary reflected adjoint. The ket runs from a to b , and the bra from b to ϑa . Both branches lie above the defect and meet with tangent $-e_0$. The scalar maps differ away from the join. Vertical scale is enlarged.

This construction removes a fixed-angle geometric and scalar-direction cusp in the direct bulk term. It does not prove absence of defect contact or junction counterterms. No driver or capacity law for (S8.2) is computed here; it is a smooth contour control for the field-theory problem, not an identification with ω .

S8.2 The specified Gaussian perturbative sector

Use the free Feynman-gauge propagators

$$\langle A_\mu^a(x) A_\nu^b(y) \rangle_0 = \frac{g^2 \delta^{ab} \delta_{\mu\nu}}{4\pi^2 |x - y|^2}, \quad \langle X_A^a(x) X_B^b(y) \rangle_0 = \frac{g^2 \delta^{ab} \delta_{AB}}{4\pi^2 |x - y|^2}. \quad (\text{S8.3})$$

This defines the direct exchange of a bulk field between the two line branches. Endpoint self-energy, endpoint-to-line interactions, defect vertices and counterterms are excluded from this sector. Only their consistent sum could be a gauge-independent interacting answer.

On a single branch the gauge/scalar numerator cancels by $M^T M = I$ or $N^T N = I$. If v_i, v_j denote oriented original ket tangents, the cross-branch numerator instead is

$$\begin{aligned} -(-Rv_i) \cdot v_j + [N(-Rv_i)] \cdot Mv_j &= v_i \cdot (R + I)v_j \\ &= 2v_{i,\text{sp}} \cdot v_{j,\text{sp}}. \end{aligned} \quad (\text{S8.4})$$

Here sp consists of the three directions unchanged by time reflection, including the defect normal. Near a regular reference join satisfying (S8.1), the spatial tangents vanish linearly in arclength. The numerator is $O(st)$ and the denominator is comparable to $(s + t)^2$. The local double integral is finite. This local estimate concerns (S8.3), not all possible defect graphs.

For the self-pairing of (S8.2), T cancels by scale invariance and the stripped cross term is

$$I(\eta) = 2\eta^2 \int_0^1 \int_0^1 \frac{f'(s)f'(t)}{(s+t)^2 + \eta^2[f(s) - f(t)]^2} ds dt. \quad (\text{S8.5})$$

If $\sum_a T_a T_a = C_F I$ in the endpoint representation, its contribution relative to the free endpoint propagator is $g^2 C_F I(\eta)/(4\pi^2)$. There is no factor $1/2$ for an exchange between two distinct ordered branches.

Proposition S8.1 (The direct bulk bulge coefficient). *For $f(t) = t^2(1 - t)$,*

$$I(\eta) = \left(\frac{3}{2} - 2 \log 2\right) \eta^2 + O(\eta^4), \quad \frac{3}{2} - 2 \log 2 = 0.1137056388801094 \dots > 0. \quad (\text{S8.6})$$

One may bound the absolute remainder by $2\eta^4$ for real η .

Proof. The quadratic coefficient is $2A$, where

$$A = \int_0^1 \int_0^1 \frac{(2s - 3s^2)(2t - 3t^2)}{(s + t)^2} ds dt. \quad (\text{S8.7})$$

Put $r = s/(s + t)$ and $u = s + t$. The Jacobian is u . By symmetry, integrate $0 \leq r \leq 1/2$ and $0 \leq u \leq 1/(1 - r)$ and multiply by two. The integrand including the Jacobian is $r(1 - r)[4 - 6u + 9r(1 - r)u^2]u$. Its u integral, including that factor two, simplifies to $r^2/[2(1 - r)^2]$. Hence

$$A = \frac{1}{2} \int_0^{1/2} \frac{r^2}{(1 - r)^2} dr = \frac{3}{4} - \log 2. \quad (\text{S8.8})$$

For the remainder, $|f'| \leq 1$, $|f'(s)| \leq 2s$ and $|f(s) - f(t)| \leq |s - t| \leq s + t$. Subtracting the denominator at $\eta = 0$ bounds the absolute integrand difference by

$$2\eta^4 \frac{|f'(s)f'(t)|[f(s) - f(t)]^2}{(s + t)^4} \leq 8\eta^4 \frac{st}{(s + t)^2} \leq 2\eta^4. \quad (\text{S8.9})$$

The integration square has area one. This justifies the expansion uniformly even at the apparent singular corner. $\square$

Numerical controls give $I(0) = 0$, $I(0.3) \simeq 0.0102220294386856$ and $I(0.7) \simeq 0.0553790171218514$. These check the integral and its small-bulge coefficient; they are not estimates of the complete interacting response.

There is also a positive Gaussian interpretation of this cross term. The reflected four-dimensional scalar covariance has representation

$$D_0(\vartheta x - y) = \int_{\mathbb{R}^3} \frac{d^3 k}{(2\pi)^3} \frac{e^{-|k|(x^0+y^0)} e^{ik \cdot (y-x)}}{2|k|}. \quad (\text{S8.10})$$

Smear it against the three spatial tangent currents of each ket and sum the components as in (S8.4). This is a Gram matrix. The vanishing of those currents at b permits the finite join limit just estimated. This proves positivity of the displayed Gaussian current sector, without extending it to the omitted diagrams.

S8.3 Why a positive norm need not be protected

For an actual reflected pairing a smooth deformation has the structure

$$\delta K_{ij} = \langle \Theta_0(\delta F_i) F_j \rangle + \langle \Theta_0 F_i \delta F_j \rangle, \quad (\text{S8.11})$$

with the insertions of Section S4 and any renormalized contact terms. Even if a deformation of a ket is Q -exact, $\delta\psi = Q\lambda$, and $Q\psi = 0$, its physical norm obeys

$$\delta \|\psi\|^2 = 2 \operatorname{Re} \langle \psi, Q\lambda \rangle = 2 \operatorname{Re} \langle Q^\dagger \psi, \lambda \rangle. \quad (\text{S8.12})$$

The first condition does not imply $Q^\dagger \psi = 0$. For example, on a Hilbert space with orthonormal e_1, e_2 , let $Qe_2 = e_1$, $Qe_1 = 0$, $\psi = e_1$, $\lambda = e_2$. Then $Q^2 = 0$, $Q\psi = 0$, but (S8.12) equals two. Adding a vector e_3 killed by both Q and $Q^\dagger$ supplies a zero-first-variation control. Thus a common-charge Ward identity and a physical positive norm are different requirements.

S8.4 The complete response still required

For an interacting candidate that survives the normalized response and localization-compatibility tests of Sections S10–S11, a further requirement is the complete first interaction-order shape response of (S7.5), beginning with a positive bulge $\eta_0 > 0$ in (S8.2). Fix the gauge, endpoint normalization, reference color sector and renormalization prescription before comparing deformations. Alongside the direct bulk exchange, include endpoint-to-line gauge exchange, the defect H-scalar derivative coupling, endpoint self-energy, and all defect and junction counterterms at that order. The interaction vertices must be taken from a consistent defect action, such as the conventions used in [4, 6]. Endpoint renormalization may cancel in a specified ratio, but that cancellation must be shown.

The flat limit needs separate analysis. At $\eta = 0$ the entire contour lies on the defect; for $\eta > 0$ only its endpoints do. Defect contact terms can therefore make the full small-height expansion nonuniform. Analyticity of the bulk integral in η^2 does not justify analyticity of the complete answer: terms linear in height or involving logarithms have not been excluded. A complete calculation at positive height would first determine whether the nonzero bulk response survives, cancels, or requires a different operator definition.

S9 Cumulative-storage gluing and joint continuation

This section gives the algebra behind the continuation criterion in the exposition [1]. It follows the inherited cumulative-storage note [19]. All operators here act on ordinary unweighted interval Hilbert spaces. The identities supply tests for a proposed construction. The final subsection reports a separate internally certified fixed-window anchor, a quantitative obstruction to a scalar coupling estimate, and a subsequent bounded append. No successful all-depth recursion is established.

S9.1 The fixed-shift spatial Schur complement

After translating I_{L+h} to $(0, L+h)$, causality gives

$$V = V_{\omega, L+h} = \begin{pmatrix} X & 0 \\ Y & Z \end{pmatrix}, \quad X = V_{\omega, L}, \quad Z = V_{\omega, h}. \quad (\text{S9.1})$$

Here Y carries old input into the added output interval. Suppose $\|X\| < 1$ and $\|Z\| < 1$, and set

$$E = I - X^*X, \quad F_{\text{out}} = I - ZZ^*, \quad C = F_{\text{out}}^{-1/2}Y E^{-1/2}.$$

These defects are coercive: each is bounded below by a positive multiple of the identity.

Proposition S9.1 (Cumulative gluing criterion). *Under these strict diagonal bounds,*

$$\|V\| \leq 1 \iff E - Y^*F_{\text{out}}^{-1}Y \succeq 0 \iff \|C\| \leq 1. \quad (\text{S9.2})$$

Replacing both norm inequalities by strict inequalities gives the strict-contraction criterion.

Proof. The defect is

$$I - V^*V = \begin{pmatrix} E - Y^*Y & -Y^*Z \\ -Z^*Y & I - Z^*Z \end{pmatrix}.$$

Its lower diagonal block is coercive. Its Schur complement is

$$E - Y^*[I + Z(I - Z^*Z)^{-1}Z^*]Y = E - Y^*F_{\text{out}}^{-1}Y = E^{1/2}(I - C^*C)E^{1/2}.$$

The middle identity follows by multiplying by $I - ZZ^*$. Schur elimination proves positivity and, using bounded invertible triangular elimination factors, equivalence of coercivity. $\square$

The output orientation $I - ZZ^*$ is essential; replacing it with $I - Z^*Z$ inside the displayed cross term changes the condition. At non-strict diagonal contraction the inverses need not exist, so this proof cannot be used without a range-compatible extension.

S9.2 Cayley coordinates and reflection

For $\omega > 0$ the finite-window transfer is Volterra convolution by a locally integrable kernel. It is quasinilpotent: conjugation by e^{-cx} gives a convolution norm bounded by $\int_0^L e^{-cu}|k_\omega(u)| du$, which tends to zero as $c \rightarrow \infty$. Each finite- c conjugation is a bounded similarity, so the spectral radius is zero. Hence $I + V$ has a bounded inverse without assuming contraction.

Define

$$\mathcal{Z} = (I - V)(I + V)^{-1}, \quad P_{\omega, L} = \frac{2}{\omega} \operatorname{Re} \mathcal{Z}, \quad \operatorname{Re} A = \frac{1}{2}(A + A^*). \quad (\text{S9.3})$$

Multiplying the real part on the two sides gives

$$I - V^*V = \frac{\omega}{2}(I + V^*)P_{\omega,L}(I + V). \quad (\text{S9.4})$$

Thus $P_{\omega,L} \succeq 0$ is exactly contraction; coercivity is exactly strict contraction in operator norm. For the real causal kernel, interval reflection $J_L f(x) = f(-x)$ on the centered interval gives $V^* = J_L V J_L$. Consequently $\mathcal{Z}^* = J_L \mathcal{Z} J_L$, and $P_{\omega,L}$ commutes with reflection. The raw input defect need not commute with it.

For the block split (S9.1), put $R_X = (I + X)^{-1}$, $R_Z = (I + Z)^{-1}$. Block inversion yields

$$P_{\omega,L+h} = \begin{pmatrix} A & B^* \\ B & F \end{pmatrix}, \quad A = P_{\omega,L}, \quad F = P_{\omega,h}, \quad B = -\frac{2}{\omega} R_Z Y R_X. \quad (\text{S9.5})$$

The two useful orientations of the diagonal identities are

$$A = \frac{2}{\omega} R_X^* E R_X, \quad F = \frac{2}{\omega} R_Z F_{\text{out}} R_Z^*. \quad (\text{S9.6})$$

Under the strict bounds of Proposition S9.1,

$$\|F^{-1/2} B A^{-1/2}\| = \|F_{\text{out}}^{-1/2} Y E^{-1/2}\|. \quad (\text{S9.7})$$

Indeed the left normalized block equals $-\mathcal{U}\mathcal{C}\mathcal{W}$, where

$$\mathcal{U} = \sqrt{2/\omega} F^{-1/2} R_Z F_{\text{out}}^{1/2}, \quad \mathcal{W} = \sqrt{2/\omega} E^{1/2} R_X A^{-1/2}.$$

The displayed diagonal identities make these invertible maps unitary. This proves (S9.7) without commuting any noncommuting factors. The Cayley change preserves the full coupling norm; it provides coordinates rather than stronger bounds.

S9.3 A step changing both length and shift

Fix the proposed append length h . At a reference positive shift α , suppose $A_\alpha = P_{\alpha,L}$ and $F_\alpha = P_{\alpha,h}$ are coercive. The enlarged block need not yet be positive. Let B_α be its cross block and put $\mathcal{K} = F_\alpha^{-1/2} B_\alpha A_\alpha^{-1/2}$. For a candidate new shift $\beta > 0$, define

$$H_A = A_\alpha^{-1/2} (A_\beta - A_\alpha) A_\alpha^{-1/2}, \quad (\text{S9.8})$$

$$H_F = F_\alpha^{-1/2} (F_\beta - F_\alpha) F_\alpha^{-1/2}, \quad (\text{S9.9})$$

$$H_B = F_\alpha^{-1/2} (B_\beta - B_\alpha) A_\alpha^{-1/2}. \quad (\text{S9.10})$$

Congruence by the reference diagonal metrics makes the enlarged form at β equal to

$$\begin{pmatrix} I + H_A & (\mathcal{K} + H_B)^* \\ \mathcal{K} + H_B & I + H_F \end{pmatrix}.$$

It is coercive if and only if $I + H_F$ is coercive and

$$\boxed{I + H_A - (\mathcal{K} + H_B)^* (I + H_F)^{-1} (\mathcal{K} + H_B) \succ 0.} \quad (\text{S9.11})$$

Here $\succ 0$ means a strictly positive operator lower bound. This is a second Schur complement; its signed changes retain improvements in either diagonal storage or cross coupling. A simple sufficient condition is

$$H_A \succeq aI, \quad H_F \succeq bI, \quad a, b > -1, \quad \|\mathcal{K} + H_B\|^2 < (1+a)(1+b).$$

Replacing each signed change by its absolute norm is more conservative and may discard the improvement that makes the step possible. All these statements concern operators, not only their finite compressions.

S9.4 Scope near zero shift and at increasing depth

The causal Cayley symbol has the core expansion

$$\frac{2}{\omega} \frac{1 - K_\omega(p)}{1 + K_\omega(p)} = a_0(p) + \omega^2 \left(\frac{a_0''(p)}{6} - \frac{a_0(p)^3}{12} \right) + O(\omega^4), \quad (\text{S9.12})$$

where primes mean p derivatives. To see the coefficient, write $\log K_\omega = -\omega a_0 - \omega^3 a_0''/6 + O(\omega^5)$ and expand the resulting hyperbolic tangent. The expression is even in ω ; its use after inversion is on appropriate smooth cores, not an operator-norm comparison with the unbounded central form. The extra cubic term also distinguishes accumulated storage from the instantaneous generator.

Positive-shift comparisons in (S9.11) avoid asserting uniform convergence on the central form domain. They still require rigorous estimates of the full blocks and their weak directions. An iteration must additionally prove $L_{j+1} = L_j + h_j$, $\sum_j h_j = \infty$ and $\omega_j \downarrow 0$. The algebra does not supply those estimates. Discarding output through a projection, assuming monotonicity in shift, or extrapolating local extension cannot replace them.

S9.5 Separate arithmetic anchor and append results

A separate computer-assisted calculation in the critical-path investigation establishes, in the original arithmetic normalization,

$$Q_{0,1/2} \succeq I/40, \quad (\text{S9.13})$$

$$D_{10^{-3},1/2} \succeq \delta I, \quad \delta = \frac{24999}{500000000} = 0.000049998.$$

The first inequality is on the central form domain; the defect bound holds for every L^2 input. The calculation uses a finite part of the fixed arithmetic exponential-moving-average (EMA) tower as a lower operator, exact endpoint response, and bounds on the entire complementary input space. Outward rational interval records passed at 40 and 60 digits [27]. This imports the stated certificate and its hypotheses, rather than reproduce its numerical proof here. Specialist mathematical review remains outstanding. It closes the earlier finite-input pilot gap and is not a new depth horizon.

For $L = 1/2$, $h = 1/20$, $\omega = 10^{-3}$, causal restriction transfers the defect floor to the smaller window. Reflection supplies the new-output orientation $F_{\text{out}} = I - ZZ^*$, so both diagonal defects in (S9.2) are bounded below by δI . The complete append response, writing old distance from the join as ρ , is

$$(Yf)(t) = \int_0^L k_\omega(t + \rho) f(L - \rho) d\rho, \quad 0 < t < h.$$

It retains the full arithmetic history, both pole contributions and the fixed local normalization.

The subsequent append calculation proves

$$55 < \frac{\|Y\|}{\delta} < 72. \quad (\text{S9.14})$$

Hence the scalar sufficient estimate $\|\mathcal{C}\| \leq \|Y\|/\delta$ cannot prove $\|\mathcal{C}\| < 1$; (S9.14) is not a lower bound above one for $\|\mathcal{C}\|$. Even after 32 cosine modes on both sides, the raw complement/complement norm divided by δ exceeds 46 [28]. Complete-output finite-input diagnostics instead give normalized quotients near 0.801830639. In exact arithmetic such finite quotients are lower, not upper, bounds on the full coupling norm. These diagnostics alone give no all-input upper bound below one.

The subsequent operator calculation retains directional defect energy and controls both full complementary spaces, establishing internally at the same (L, h, ω)

$$\left\| F_{\text{out}}^{-1/2} Y E^{-1/2} \right\| < 0.951 \quad [29].$$

It uses a forward/backward energy factorization and a Carleman operator bound for the singular corner, preserving the complete history and the output-defect orientation. Its proof and rational records remain in the critical-path investigation; independent specialist review is outstanding. It establishes neither an all-depth continuation nor a physical realization. No instantaneous operator lower bound is exponentiated through the nonnormal flow. Auxiliary smoothing loss and averaging variance are not arithmetic positivity; the field resolution in Section S13 likewise makes no contribution to this arithmetic estimate.

S10 A normalized Wilson response on a fixed window

This section tests a definite response on $0 < L < \log 2$. It gives the full arithmetic comparison, a regulated Wilson readout and its Gaussian first variation, followed by two scoped obstructions. The calculation does not construct the effective transfer hypothesized in the exposition.

S10.1 The causal target and its fixed contact term

For $f \in C_c^\infty(I_L)$ let F be its zero extension and $d = x + L/2$. No nontrivial integer delay acts on this window. The local constant and pole terms in (S1.13) still give

$$\begin{aligned} (G_{0,L}f)(x) &= w_0 f(x) + 2 \int_0^\infty n_\Gamma(u) \{f(x) - F(x-u)\} du \\ &\quad + 4 \int_0^d \cosh(u/2) f(x-u) du. \end{aligned} \quad (\text{S10.1})$$

This is the nonsymmetric causal generator. The symmetric pole kernel is $2 \cosh((x-y)/2)$, with form $2|C(f)|^2 - 2|S(f)|^2$, and must not be substituted for the causal pole kernel.

The gamma tail outside the available delay interval is local but does not vanish. Substituting $v = e^{-u/2}$ gives

$$T(d) := 2 \int_d^\infty n_\Gamma(u) du = \log \frac{1 + e^{-d/2}}{1 - e^{-d/2}} + 2 \arctan(e^{-d/2}). \quad (\text{S10.2})$$

Thus the first two terms in (S10.1) are $(w_0 + T(d))f(x) + 2 \int_0^d n_\Gamma(u) \{f(x) - f(x-u)\} du$. Deleting the tail would change the target.

Proposition S10.1 (Full first-response distribution). *Set $A_L = -G_{0,L}$ and, for $u > 0$,*

$$b(u) = 2n_\Gamma(u) - 4 \cosh(u/2) = \frac{2e^{-5u/2}}{1 - e^{-2u}} - 2e^{u/2}. \quad (\text{S10.3})$$

With $c_\epsilon = -w_0 - T(\epsilon)$, one has on the smooth core, in $L^2(I_L)$,

$$(A_L f)(x) = \lim_{\epsilon \downarrow 0} \left[c_\epsilon f(x) + \int_\epsilon^d b(u) f(x-u) du \right], \quad (\text{S10.4})$$

where the integral is zero for $d \leq \epsilon$. Moreover

$$c_\epsilon = \log \epsilon + \gamma + \log(2\pi) + O(\epsilon). \quad (\text{S10.5})$$

Proof. The gamma part of the bracket is $-w_0 f - 2 \int_{\epsilon}^{\infty} n_{\Gamma}(u)(f - F(\cdot - u)) du$. Its omitted part tends to zero in L^2 because $\|F - F(\cdot - u)\|_2 \leq u \|F'\|_2$ and $n_{\Gamma}(u) = O(1/u)$ near zero. The omitted pole integral is $O(\epsilon) \|f\|_2$ by the convolution bound. Finally, (S10.2) gives $T(\epsilon) = -\log \epsilon + \log 4 + \pi/2 + O(\epsilon)$. Insert the specified $w_0 = -\gamma - \pi/2 - 3 \log 2 - \log \pi$. $\square$

The finite part in (S10.5) belongs to this hard-delay-cutoff convention. A different physical regulator needs an explicit conversion to it; no freedom to retune the arithmetic constant is introduced. The off-diagonal sign also matters. With $z = e^u$, $b(u) = 0$ is $z^3 - z - 1 = 0$, whose unique root with $z > 1$ gives $u_* = 0.2811995743229617 \dots < \log 2$. The response kernel is positive below u_* and negative above it. This excludes a pointwise nonnegative first-response kernel on a window longer than u_* ; it does not exclude positive Gram operators, whose entries or derivatives need not be pointwise positive.

S10.2 The lowest-mode cancellation at finite shift

The prime-free arithmetic symbol is

$$a_0^<(p) = \psi\left(\frac{1}{4} + \frac{p}{2}\right) - \log \pi + \frac{2}{p+1/2} + \frac{2}{p-1/2}.$$

The digamma recurrence [24, Section 5.5] gives

$$a_0^<(p) = \psi\left(\frac{5}{4} + \frac{p}{2}\right) - \log \pi + \frac{2}{p-1/2}. \quad (\text{S10.6})$$

Equivalently, with $n_1(u) = e^{-2u} n_{\Gamma}(u)$ and $w_1 = w_0 + 4$,

$$G_{0,L} = w_1 I + 2 \int_0^{\infty} n_1(u)(I - T_u) du + 2R_{-1/2,L}, \quad (R_{-1/2,L}f)(x) = \int_0^d e^{u/2} f(x-u) du. \quad (\text{S10.7})$$

The shift in the local term exactly accounts for removing the $n = 0$ gamma mode. The remaining growing exponential is bounded as a Volterra kernel on this fixed interval; it is not a decaying covariance mode.

Absorbing $(p+1/2-\omega)/(p+1/2+\omega)$ into the gamma ratio in (S1.16) yields, for $0 < \omega \leq 1/2$,

$$K_{\omega}^<(p) = \pi^{\omega} \frac{\Gamma((p+5/2-\omega)/2)}{\Gamma((p+5/2+\omega)/2)} \left(1 - \frac{2\omega}{p-1/2+\omega}\right). \quad (\text{S10.8})$$

Its compression equals $V_{\omega,L}$ on $L < \log 2$. It is not the full whole-line zeta quotient. In particular the apparent pole of this factor must not be interpreted as a pole of the entire completed-zeta function.

Define

$$h_{\omega}(u) = \frac{2\pi^{\omega}}{\Gamma(\omega)} e^{-(5/2-\omega)u} (1 - e^{-2u})^{\omega-1} \mathbf{1}_{u>0}. \quad (\text{S10.9})$$

Inverse Laplace transformation on $\text{Re } p > 1$ gives

$$k_{\omega}^<(u) = h_{\omega}(u) - 2\omega \int_0^u e^{(1/2-\omega)(u-v)} h_{\omega}(v) dv. \quad (\text{S10.10})$$

It is locally integrable at every positive shift, with leading singularity proportional to $u^{\omega-1}$. Its identity limit at zero is distributional. Differentiating the corresponding operators on the smooth core recovers (S10.7), including the pole.

S10.3 A regulated readout with identity initial data

For a fully specified control, take independent abelian Gaussian bulk gauge components and six Hermitian scalars with equal Feynman-gauge covariance $g^2/(4\pi^2|X - Y|^2)$. Take an independent complex defect doublet with covariance $\delta_{mn}/|a - b|$. This is a fixed-gauge Gaussian probe model, not the complete interacting gauge/defect action. An explicit regulator uses heat times $s \geq \epsilon^2$:

$$D_\epsilon(X - Y) = g^2 \int_{\epsilon^2}^{\infty} (4\pi s)^{-2} e^{-|X - Y|^2/(4s)} ds, \quad (\text{S10.11})$$

$$C_\epsilon(a - b) = 4\pi \int_{\epsilon^2}^{\infty} (4\pi s)^{-3/2} e^{-|a - b|^2/(4s)} ds. \quad (\text{S10.12})$$

The probability measure is normalized and independent of the deformation. Reflection positivity of this cutoff measure is not assumed.

Use the tangent map M of Section S5, with columns $(-e_7, e_8, e_9, e_6)$, and the curves

$$X_{r,\eta}(t) = (rt, \eta r \varphi(t)), \quad \varphi(t) = t^2(1 - t), \quad 0 \leq t \leq 1.$$

The ket runs from $t = 1$ to zero, ending at the common reference $b = 0$ with tangent $-e_0$. Let $r = e^x$ and $\eta(\omega) = \eta_0 + \omega$, $\eta_0 > 0$. This is a test deformation, not an identification with Loewner capacity. The color fiber is one-dimensional with its ordinary metric and an identity contraction at b ; no extra defect field is inserted there.

Use the physical pair $q_2, \bar{q}_2$, and define

$$F_{\omega,\epsilon}(x) = e^{x/2} U_M(C_{e^x, \eta(\omega)}) q_{2,\epsilon}(a_{e^x}), \quad (\text{S10.13})$$

$$K_{\omega,\epsilon}(x, y) = \langle \Theta_0 F_{\omega,\epsilon}(x) F_{\omega,\epsilon}(y) \rangle_\epsilon. \quad (\text{S10.14})$$

The bra uses reversed ordering and $N = -MR$, as in Section S7. Replacing $\bar{q}_2$ by the same-charge $\bar{q}_1$ would give the vanishing endpoint control of Section S6. No common endpoint supercharge or protected expectation is asserted.

Specify the response on every input by

$$(\mathcal{V}_{\omega,L}^\epsilon f)(x) = f(x) + \int_{-L/2}^x [K_{\omega,\epsilon}(x, y) - K_{0,\epsilon}(x, y)] f(y) dy. \quad (\text{S10.15})$$

The integral pairs the output row with the source state $\int_{-L/2}^x F_\omega(y) f(y) dy$, minus its baseline. This supplies a direct identity channel and exact $\mathcal{V}_{0,L}^\epsilon = I$, without inverting a compact Gram operator. Causality in x is imposed by the readout, not derived from Euclidean reflection. The adjoint in the fixed unweighted norm is the usual upper-triangular integral adjoint, with conjugated kernel. It is not contour reversal with unchanged scalar map.

S10.4 A nonzero but bounded Gaussian first variation

The exact fixed-endpoint tangent-connection insertion (S4.11) applies. Its curvature contains the scalar and polarization terms of (S4.4). The bulge displacement vanishes at both endpoints; their polarizations and radial weights are fixed. Both bra and ket vary. More generally one would have

$$\partial_\omega \langle \mathcal{O} \rangle_\omega = \langle \partial_\omega \mathcal{O} \rangle_\omega - \langle \mathcal{O} \partial_\omega S_\omega \rangle_{\omega,c},$$

as well as derivatives of external normalizations or source identifications. The second term is zero in this model because its normalized measure is fixed.

At free endpoint order the kernel is $G_o(x - y) = 1/(2 \cosh((x - y)/2))$, independent of the bulge, and the response is I . The Wilson dressing gives a nonzero control. Same-branch gauge/scalar contractions cancel. The ordinary reflected cross contraction gives, in the continuum abelian Gaussian model,

$$K_\omega(x, y) = G_o(x - y) \exp\{\lambda B_{\eta(\omega)}(r, s)\}, \quad r = e^x, \quad s = e^y, \quad \lambda = \frac{g^2}{4\pi^2}, \quad (\text{S10.16})$$

where

$$B_\eta(r, s) = 2\eta^2 rs \int_0^1 \int_0^1 \frac{\varphi'(v)\varphi'(t)}{(rv + st)^2 + \eta^2[r\varphi(v) - s\varphi(t)]^2} dv dt. \quad (\text{S10.17})$$

The exponential is exact only for these independent abelian Gaussian fields. Its first order in g^2 is the direct bulk sector of Section S8. It is not the sum of the interacting endpoint, defect and counterterm diagrams.

Set $A = (rv + st)^2$ and $H = [r\varphi(v) - s\varphi(t)]^2$. Direct differentiation gives

$$\partial_\eta B_\eta(r, s) = 4\eta rs \int_0^1 \int_0^1 \frac{\varphi'(v)\varphi'(t)A}{(A + \eta^2 H)^2} dv dt. \quad (\text{S10.18})$$

Using $|\varphi'(v)| \leq 2v$ and $4rsvt \leq (rv + st)^2$ gives

$$|B_\eta(r, s)| \leq 2\eta^2, \quad |\partial_\eta B_\eta(r, s)| \leq 4\eta. \quad (\text{S10.19})$$

These estimates include the reference corner $v = t = 0$. For the regulator above, its four-dimensional covariance is a constant times $(1 - e^{-\rho^2/(4\epsilon^2)})/\rho^2$. It and its radial derivative are bounded by constant multiples of ρ^{-2} and ρ^{-3} , uniformly in ϵ . On compact positive ranges of r, s, η , the displacement of the separation is $O(v + t)$. The differentiated cross integrands are consequently bounded by a constant times $vt/(v + t)^2$. Dominated convergence justifies the continuum kernel and its first derivative in this model. It does not justify removal of interacting defect or junction counterterms.

At $\omega = 0$,

$$J_0(x, y) = \lambda G_o(x - y) e^{\lambda B_{\eta_0}(r, s)} \partial_\eta B_{\eta_0}(r, s), \quad |J_0| \leq 2\lambda\eta_0 e^{2\lambda\eta_0^2}. \quad (\text{S10.20})$$

Uniform first differentiability on the finite square therefore gives

$$\mathcal{V}_{\omega, L} = I + \omega \mathcal{J}_L + o(\omega) \quad \text{in operator norm}, \quad \|\mathcal{J}_L\| \leq 2L\lambda\eta_0 e^{2\lambda\eta_0^2}, \quad (\text{S10.21})$$

where $\mathcal{J}_L$ is the lower-triangular integral operator with kernel J_0 . For example, $B_{0.3}(1, 1) = 0.0102220294386856\dots$ and $\partial_\eta B_{0.3}(1, 1) = 0.0680705213083742\dots$. The derivative is nonzero but finite at $x = y$.

S10.5 Two obstructions and their precise scope

Proposition S10.2 (Bounded first derivatives cannot match). *If a response has first derivative on the smooth core equal to the restriction of a bounded $L^2(I_L)$ operator, it cannot equal $-G_{0,L}$ there. In particular this excludes (S10.21), even after adding a bounded local multiplier to its derivative.*

Proof. Fix $0 \neq \chi \in C_c^\infty(I_L)$ and set $f_N = e^{iN\chi}$. From (S1.4) and the digamma asymptotic [24, Section 5.11],

$$\frac{Q_{0,L}[f_N]}{\|\chi\|^2} = \log N - \log(2\pi) + o(1). \quad (\text{S10.22})$$

Indeed the Fourier transform is translated by N ; split the integral into its central region and the rapidly decaying tails of $\hat{\chi}$. The two pole moments tend to zero by integration by parts. If a bounded A equaled $-G_{0,L}$ on the core, then $\text{Re} \langle f_N, Af_N \rangle / \|\chi\|^2$ would tend to $-\infty$, contrary to boundedness of A . $\square$

This uses the localized form to disprove full operator equality, not to weaken the matching target. It does not assign a whole-line unweighted Fourier multiplier to the growing pole. A finite smooth clock change still has a bounded derivative. A singular clock or a regulator limit with unbounded first-derivative norms requires a separate analysis and is not excluded. In particular smoothness at each fixed cutoff alone is not a continuum obstruction.

Proposition S10.3 (A persistent identity channel cannot match). *For every $\omega > 0$, $V_{\omega,L}$ is compact. A response of the form $I + C_\omega$ with compact C_ω satisfies*

$$\|I + C_\omega - V_{\omega,L}\| \geq 1.$$

The Gaussian readout (S10.15) has this form.

Proof. The kernel (S10.10) is in $L^1(0,L)$. Approximate it there by bounded kernels. Young's convolution bound gives convergence in operator norm, while every approximating operator on the finite square is Hilbert–Schmidt. Hence $V_{\omega,L}$ is compact. The difference in the proposition is identity plus compact, whose essential norm is one. The Gaussian kernel difference is bounded on the finite square, so its retarded operator is Hilbert–Schmidt. $\square$

A nonzero scalar direct coefficient gives the lower bound $|m_\omega|$ instead. More generally a nonzero multiplication operator on this nonatomic interval is not compact. There is no contradiction with the arithmetic strong initial limit: compact operators can converge strongly to I , but not in operator norm. A successful response must supply the singularly concentrating identity limit rather than retain a separate regular direct channel.

S10.6 Consequences for storage and continuation

Neither a nonzero Gaussian variation nor a positive reflected Gram kernel gives the physical balance law $\|f\|^2 = \|\mathcal{V}_{\omega,L}f\|^2 + \|\mathcal{B}_{\omega,L}f\|^2$. After transfer matching, that law requires $\|\mathcal{B}_{\omega,L}f\|^2 = 2\omega Q_{0,L}[f] + o(\omega)$. In a direction with $Q_{0,L}[f] > 0$, the amplitude is of order $\sqrt{\omega}$. A state difference that is $O(\omega)$ in its positive Hilbert norm cannot supply this first-order defect. This statement is conditional on that norm regularity.

The cumulative defect need not be monotone in shift; no positivity of every instantaneous generator is imposed here. The cross-interval bound using the new slab's output defect and the signed shift changes in Section S9 remain separate tasks. Nor does this prime-free test account for the first new generator delay $-2(\log 2)/\sqrt{2}T_{\log 2}$ when $\log 2 < L < \log 3$. The excluded regular readout and the unconstructed arithmetic Loewner driver must not be conflated with all possible Wilson realizations.

S11 Localization and an auxiliary boson–fermion product

Localization is a candidate explanation of the cancellations in Section S10, following Edward Baker's suggestion. This section gives an exact auxiliary product and a test for its possible physical origin. It does not derive the product from a localization complex.

S11.1 The defect-compatible localization question

Pestun's spherical Wilson-loop calculation organizes fluctuation cancellations through equivariant kernel and cokernel multiplicities [22, Section 4]. Baker's open-line discussion instead anticipates reduction to a one- or two-dimensional defect theory [6, Section 3.3]. Wang's framework couples a two-dimensional gauge theory to boundary quantum mechanics and includes bilocals with endpoint matter and gauge transport [23, Section 5.1, equation (5.5)]. The normal-fluctuation determinant is explicitly left unevaluated in [23, Section 4.4]. These results provide a framework to examine, not the determinant or boundary response required here.

The conformal open semicircle is an appropriate control. The constant-Poincare-charge endpoint restriction of Section S6 does not exclude a conformal Killing-spinor construction. Localizing the whole observable requires its Q -closure and boundary conditions compatible with the bosonic symmetry generated by Q^2 . Closure of each factor separately is sufficient, but is not assumed to be the only possibility.

S11.2 An exact product for the complete prime-free factor

Retain one complex bosonic Gaussian at each $a_n = 2n + 1/2$, $n = 1, \dots, N$, and one complex Grassmann pair of weight $p - 1/2$. For real $p > 1$ define

$$\begin{aligned} Z_N(p) &= \left(\int d\bar{\chi} d\chi e^{-(p-1/2)\bar{\chi}\chi} \right) \prod_{n=1}^N \left(\int_{\mathbb{C}} \frac{d^2 z_n}{\pi} e^{-(p+a_n)|z_n|^2} \right) \\ &= (p - 1/2) \prod_{n=1}^N (p + a_n)^{-1}. \end{aligned} \quad (\text{S11.1})$$

The Berezin orientation is chosen to give the factor $p - 1/2$. The bosonic integrals converge also at $p \pm \omega$ for $0 \leq \omega \leq 1/2$. One real Gaussian would give a square root, so statistics and multiplicities are substantive data.

Proposition S11.1 (Auxiliary full-factor identity). *For $\text{Re } p > 1$ and $0 \leq \omega \leq 1/2$,*

$$\pi^\omega \lim_{N \rightarrow \infty} N^{-\omega} \frac{Z_N(p - \omega)}{Z_N(p + \omega)} = K_\omega^<(p), \quad (\text{S11.2})$$

with $K_\omega^<$ as in (S10.8).

Proof. Put $z_\pm = (p + 1/2 \pm \omega)/2$. The finite product is

$$\frac{Z_N(p - \omega)}{Z_N(p + \omega)} = \frac{p - 1/2 - \omega}{p - 1/2 + \omega} \frac{\Gamma(N + 1 + z_+) \Gamma(1 + z_-)}{\Gamma(N + 1 + z_-) \Gamma(1 + z_+)}. \quad (\text{S11.3})$$

The first gamma ratio in N is asymptotic to N^ω . Multiplication by $N^{-\omega} \pi^\omega$ proves the result. The real Gaussian identity extends analytically to the stated right half-plane. $\square$

Removing the $n = 0$ boson from the earlier gamma model accounts for the first rational cancellation. The surviving Grassmann factor supplies the remaining pole ratio. This is an auxiliary identity with modes prescribed from the arithmetic target. The coexistence of bosons and fermions does not specify a supersymmetric action or prove their pairing by a physical charge. The Berezin factor is also not a positive Hilbert-space norm.

The normalization in (S11.2) remains external. In fact, differentiating its finite expression gives

$$-\partial_\omega \log \left[\pi^\omega N^{-\omega} \frac{Z_N(p-\omega)}{Z_N(p+\omega)} \right] \Big|_{\omega=0} = \log N - \log \pi + \frac{2}{p-1/2} - 2 \sum_{n=1}^N \frac{1}{p+a_n}. \quad (\text{S11.4})$$

The limit is precisely (S10.6). Reinstating the removed mode gives the original w_0 , or equivalently the fixed finite part of (S10.5). A physical finite-renormalization choice must reproduce it; counterterm freedom alone does not explain its value.

S11.3 A mode-counting test and the remaining dictionary

Differentiate (S10.6) in p to remove an additive p -independent generator constant:

$$\begin{aligned} \partial_p a_0^<(p) &= 2 \sum_{n=1}^{\infty} \frac{1}{(p+2n+1/2)^2} - \frac{2}{(p-1/2)^2} \\ &= 2 \int_0^{\infty} u e^{-pu} \left[\frac{e^{-5u/2}}{1-e^{-2u}} - e^{u/2} \right] du. \end{aligned} \quad (\text{S11.5})$$

Both expressions converge for $\text{Re } p > 1$. They provide a sharp comparison for a computed virtual mode character, but do not remove arbitrary p -dependent counterterms. The relative weight $-1/2$ is not by itself a negative physical energy: the auxiliary integral depends on $p-1/2$ on a right half-plane. The interpretation of p as a physical boundary or equivariant parameter is missing.

A viable localization calculation must determine the charge, endpoint conditions, fluctuation complex, boundary modes and surviving multiplicities, rather than insert (S11.5). The reduced theory's correlators and the observable can contribute alongside its one-loop determinant; the required factor need not all arise from that determinant. An explicit preserved charge and the resulting mode contribution, or a scoped obstruction, would be a useful first deliverable.

The physical shift also needs its own definition. Under the localization hypotheses a fixed Q -closed expectation is independent of the coefficient of a Q -exact localization term. That coefficient cannot simply become ω . A supersymmetric mass, boundary coupling, admissible contour parameter or source deformation may vary nontrivially, but its dictionary and all measure and observable variations must be calculated.

Finally (S11.2) is a partition ratio. A source-to-response construction must still produce causal support on arbitrary boundary inputs, the correct identity limit, and the ordinary physical adjoint. If it retains the regular readout of (S10.15), Propositions S10.2 and S10.3 still apply. Signed cohomological cancellation cannot be substituted for the independent positive cumulative norm. Spatial gluing, the integer delays beyond $\log 2$, and contraction on a sequence $L_j \rightarrow \infty$, $\omega_j \downarrow 0$ remain open.

The independently specified hemisphere and scalar-endpoint control is calculated in Section S12. It supplies a one-sided gamma amplitude, but its fixed protected representation fails ordinary spatial descent. Its canonical-Hodge alternative and the distinction between a composite-state character and an endpoint response are also tested there.

S12 Localized boundary amplitudes and spatial radial evolution

The auxiliary product in Section S11 prescribes its modes from arithmetic. We now test an independently specified free localization problem. It supplies a one-sided gamma amplitude and an explicit scalar-endpoint state, but no identification with the spatial transfer. The obstruction

occurs before fitting the pole factors: ordinary spatial dilation fails to preserve the fixed protected representatives. Canonical Hodge representatives allow a different radial evolution, whose elementary scalar channel contains only the lowest spatial mode.

The localization and Hodge inputs are established results of [25, 26]. Their application and the comparison below are internal research calculations, without independent specialist review. The statements concern the specified free sector, not every supersymmetric defect or Wilson construction.

S12.1 A specified charge, boundary state and circle control

Take one free three-dimensional $\mathcal{N} = 4$ hypermultiplet on a round hemisphere HS^3 of radius r . Use the Higgs boundary polarization of [26, Sections 4.2 and 4.3.2]. In that reference's sphere-algebra notation the charge is

$$\mathcal{Q}^H = Q_1^+ + Q_2^- = \mathcal{Q}_1^{\ell+} + \mathcal{Q}_1^{r-} + \mathcal{Q}_2^{\ell-} + \mathcal{Q}_2^{r+}. \quad (\text{S12.1})$$

The ℓ, r superscripts here label supersymmetry algebras. They do not denote spatial angular momentum or an additional radius. For this equivariant charge, the protected complex is understood on the $\mathcal{Q}^2$ -invariant subspace; unrestricted nilpotence is not assumed. The scalar boundary data and auxiliary completion are

$$q_1| = Y, \quad \tilde{q}_2| = \bar{Y}, \quad G| = -\frac{i}{2r}Y, \quad \bar{G}| = \frac{i}{2r}\bar{Y}, \quad (\text{S12.2})$$

with the fermionic completion in [26, equations (80)–(82)]. In the free theory, $G = i\partial_\perp q_2$, so $\partial_\perp q_2| = -Y/(2r)$; zero Neumann data would define a different problem. There is no dynamical gauge multiplet in this control. An embedding into the four-dimensional Wilson problem is not assumed.

Put $\epsilon = (4\pi r)^{-1}$ and $x = |Y|^2/\epsilon$. The localized empty-hemisphere state is

$$\Psi_0(Y, \bar{Y}) = (\pi\epsilon)^{-1/2}e^{-x}, \quad \int_{\mathbb{C}} |\Psi_0|^2 d^2Y = \frac{1}{2}. \quad (\text{S12.3})$$

For a neutral radial state $\Psi = (\pi\epsilon)^{-1/2}f(x)$, the normalized boundary Fourier–Mellin transform has radial part

$$\mathcal{M}f(\sigma) = \frac{1}{\sqrt{2\pi}} \int_0^\infty x^{s-1} f(x) dx, \quad s = \frac{1}{2} - i\sigma, \quad \sigma \in \mathbb{R}. \quad (\text{S12.4})$$

The angular Fourier label is zero for these states. In particular,

$$\mathcal{M}(e^{-x})(\sigma) = \frac{\Gamma(s)}{\sqrt{2\pi}}. \quad (\text{S12.5})$$

The integral converges for $\text{Re } s > 0$ and extends meromorphically. Its one-sided simple poles are therefore a prediction of this specified boundary amplitude, without an arithmetic mode prescription. This agrees with the mirror hemisphere amplitude in [26, equations (135)–(140)].

The circle determinant is a different object. The polarized scalar fields have twist $u^a(\varphi) = (\cos(\varphi/2), \sin(\varphi/2))$ and antiperiodic boundary conditions. At fixed real mass/background μ , their kinetic operator is $\partial_\varphi + \mu$ and has weights $\mu + i(n + 1/2)$, $n \in \mathbb{Z}$ [25, Section 6.1]. Pairing both frequency directions gives

$$\frac{Z_{\text{circ}}(\mu)}{Z_{\text{circ}}(0)} = \prod_{n=0}^\infty \frac{(n + 1/2)^2}{(n + 1/2)^2 + \mu^2} = \frac{1}{\cosh \pi\mu}, \quad Z_{\text{circ}}(0) = \frac{1}{2}. \quad (\text{S12.6})$$

Correspondingly, $|\Gamma(1/2 - i\sigma)|^2/(2\pi) = [2 \cosh(\pi\sigma)]^{-1}$. The gamma amplitude before gluing and the paired determinant must not be interchanged. In particular, the direct real-mass shift ratio tends to $e^{2\pi\omega}$ at large mass, whereas the arithmetic factor (S10.8) behaves as $(2\pi/p)^\omega$ at large positive p .

S12.2 The actual two-scalar state and its adjoint

The boundary operators act by

$$Q_L = Y, \quad \tilde{Q}_L = -\epsilon \partial_Y, \quad \tilde{Q}_R = \bar{Y}, \quad Q_R = -\epsilon \partial_{\bar{Y}}, \quad [\tilde{Q}_L, Q_L] = -\epsilon I. \quad (\text{S12.7})$$

The right operators use the opposite ordering of the right action [26, equations (125)–(126)]. For the free massless semicircle, separated opposite-pole scalar endpoints with identity transport give

$$\Psi_{\text{pair}} = Q_L \tilde{Q}_R \Psi_0 = Y \bar{Y} \Psi_0 = \epsilon x \Psi_0, \quad (\text{S12.8})$$

$$\hat{\Psi}_{\text{pair}} = \frac{\epsilon}{\sqrt{2\pi}} \Gamma(s+1). \quad (\text{S12.9})$$

The normalized gluing overlap is $\epsilon/2$, agreeing with the ordered propagator $-\epsilon \operatorname{sgn}(\varphi_1 - \varphi_2)/2$. No contact subtraction is forced at separated endpoints. Reversing the coincident ordering gives $\epsilon(x-1)\Psi_0$, and choosing the symmetric local composite gives $\epsilon(x-1/2)\Psi_0$. These are distinct observables; none forces an additional quartic insertion into (S12.8).

The real-coordinate boundary norm reduces to $L^2((0, \infty), dx)$. For $N = -x\partial_x$, integration by parts and Mellin Plancherel give

$$N^* = 1 - N, \quad \mathcal{M}(Nf) = s \mathcal{M}f, \quad \|f\|_{L^2(dx)}^2 = \int_{\mathbb{R}} |\mathcal{M}f(\sigma)|^2 d\sigma. \quad (\text{S12.10})$$

These identities hold first on a suitable smooth core. The centered operator $N - 1/2$ is skew-adjoint and generates unitary dilation of the *field magnitude*. It has not been identified with spatial dilation. On the real contour ordinary conjugation sends s to $1 - s$. For example,

$$\int_0^\infty |xe^{-x}|^2 dx = \frac{1}{4}, \quad \int_{\mathbb{R}} \frac{|s|^2}{2 \cosh \pi \sigma} d\sigma = \frac{1}{4}, \quad \int_{\mathbb{R}} \frac{s^2}{2 \cosh \pi \sigma} d\sigma = 0.$$

Reusing a vacuum gluing phase with the same analytic polynomial on both sides would make the last, incorrect norm substitution. This boundary-coordinate identity also does not prove an isometry from the full physical radial state space or arithmetic storage.

S12.3 Why ordinary spatial dilation fails on fixed representatives

The hemisphere cut is at great-circle angles $\varphi = 0, \pi$. For concentric radial slices, stereographically project from the antipode of its center. On the protected diameter,

$$t = 2r \tan \frac{\varphi - \pi/2}{2}, \quad \Omega = \left(1 + \frac{t^2}{4r^2}\right)^{-1}, \quad \frac{u(\varphi)}{\sqrt{\Omega}} = \frac{1}{\sqrt{2}} \left(1 - \frac{t}{2r}, 1 + \frac{t}{2r}\right). \quad (\text{S12.11})$$

The hemisphere is the ball of radius $2r$. A constant H-basis rotation writes the protected scalar as

$$O(t) = q'_1(t) + \frac{t}{2r} q'_2(t), \quad \mathcal{Q}q'_2(t) = \chi(t), \quad \mathcal{Q}q'_1(t) = -\frac{t}{2r} \chi(t). \quad (\text{S12.12})$$

Here χ is a nonzero free-fermion combination. These relations are the affine conformal Killing-spinor variation underlying the twist in [25, equations (3.1)–(3.5)]. The basis change is passive and does not replace (S12.2).

Proposition S12.1 (Failure on fixed protected representatives). *At fixed finite r , ordinary spatial dilation D does not preserve all closed representatives of the specified free Higgs complex. In particular it does not directly induce the field-space Mellin evolution (S12.10).*

Proof. The scalar has conformal dimension $1/2$. Dilation acts on the fields, keeping the coefficients of the specified insertion fixed. Thus

$$[D, O(t)] = (t\partial_t + \tfrac{1}{2})O(t) - \frac{t}{2r}q'_2(t), \quad \mathcal{Q}[D, O(t)] = -\frac{t}{2r}\chi(t). \quad (\text{S12.13})$$

The derivative differentiates the complete protected family. The last expression gives a nonzero witness at a nonzero interior point. Equivalently, finite dilation leaves the residual $(1 - e^a)tq'_2(e^at)/(2r)$, apart from the common factor $e^{a/2}$. Acting on the conformal vacuum gives the same boundary-state counterexample. A neutral separated pair has two such fermionic terms; setting one insertion at $t = 0$ leaves the other witness. Hence the failure is preservation of closed representatives, not merely a nonzero commutator between D and $\mathcal{Q}$. $\square$

The compensation $\hat{D} = D - R_H$, with adapted H weights $(1/2, -1/2)$ on the doublet, satisfies $[\hat{D}, O(t)] = t\partial_t O(t)$ and is the exact twisted dilation on the relevant local cohomology [25, equation (3.4)]. It acts trivially there. The corresponding free protected two-point function is proportional to an ordering sign: the polarization determinant cancels $|t_1 - t_2|^{-1}$. It does not retain the stationary untwisted spatial covariance of Section S2.

Transporting the charge itself instead relates different complexes: in the standard flat frame, $e^{aD}\mathcal{Q}_re^{-aD} = e^{a/2}\mathcal{Q}_{e^ar}$. That alternative needs specified intertwiners and ordinary adjoints. Proposition S12.1 does not exclude it, an enlarged unprotected state space, another supersymmetric sector, or the canonical representative construction below.

S12.4 Canonical Hodge evolution and the lost spatial tower

For either nilpotent Higgs charge in the free conformal theory, the positive Hodge operator is proportional to $B_H = D - R_H \succeq 0$. Its kernel gives the common canonical representatives [25, equation (3.4)]. In the joint energy/R-charge basis, D commutes with $P_H = 1_{\{0\}}(B_H)$. Consequently

$$T_H(u) = e^{-uD}|_{\ker B_H}, \quad u \geq 0, \quad (\text{S12.14})$$

is a contraction semigroup and $D = R_H$ on this space. Choosing these representatives first is a different operation from the failed descent on arbitrary representatives. It does not make D zero or prove an isometry with the hemisphere coordinate norm.

Proposition S12.2 (Elementary spatial channel after Hodge projection). *For a highest-H-weight elementary free scalar, Hodge projection retains only its lowest spatial harmonic. In the parity-even axial covariance channel, the retained and omitted terms are*

$$C_H(u) = e^{-u/2}, \quad C_{\text{omit}}(u) = \frac{e^{-5u/2}}{1 - e^{-2u}}, \quad u > 0. \quad (\text{S12.15})$$

Proof. The conformally rescaled scalar cylinder operator is $-\partial_u^2 - \Delta_{S^2} + 1/4$. On a one-particle harmonic,

$$D = \ell + \tfrac{1}{2}, \quad R_H = \tfrac{1}{2}, \quad B_H = \ell.$$

Only $\ell = 0$ lies in the Hodge kernel. The parity-even observable channel in Section S2 is

$$C_{\text{ev}}(u) = \tfrac{1}{2}[C(u, 1) + C(u, -1)] = \sum_{n \geq 0} e^{-(2n+1/2)u} = \frac{e^{-u/2}}{1 - e^{-2u}}.$$

Subtracting its $\ell = 0$ term proves (S12.15). $\square$

The full spatial multiplicity remains $2\ell + 1$; the unit coefficients above describe the selected channel. The aligned and antipodal observations are used at $u > 0$, not as a normalized vector at $u = 0$. Their omitted fraction is e^{-2u} , tending to one as $u \downarrow 0$, and, for $\text{Re } p > 1/2$ in the arithmetic comparison,

$$C_{\text{omit}}(u) = \frac{1}{2u} - \frac{3}{4} + O(u), \quad 2 \int_0^\infty u e^{-pu} C_{\text{omit}}(u) du = 2 \sum_{n \geq 1} \frac{1}{(p + 2n + 1/2)^2}. \quad (\text{S12.16})$$

This is exactly the shape of the positive tower in (S11.5). The full arithmetic derivative still contains $-2/(p - 1/2)^2$. Equation (S12.16) does not count the discarded tower as arithmetic positivity. In particular, removing the lowest scalar mode after this projection leaves no elementary spatial tower at all.

S12.5 Composite-state characters and endpoint spectral weights

The many-particle protected space contains more states. Let A, B denote independent highest-H-weight scalar creators of flavor charges $+1, -1$, not ordinary adjoints. The free Higgs chiral ring is polynomial in these generators [26, Section 4.3.2]. Normalized monomials $A^a B^b$ have energy $(a + b)/2$ and flavor $a - b$, so

$$\chi_H(u, y) = \sum_{a, b \geq 0} e^{-u(a+b)/2} y^{a-b} = \frac{1}{(1 - ye^{-u/2})(1 - y^{-1}e^{-u/2})}. \quad (\text{S12.17})$$

At flavor one, the states $A^{k+1} B^k$ have energies $k + 1/2$. Selecting even k gives the ladder $2n + 1/2$; additionally deleting the ground state gives the character C_{omit} . These are additional selections of composite states, not the physical single-particle angular projection.

For an elementary normalized endpoint ket $E|0\rangle = |1, 0\rangle$, the actual ordinary-adjoint radial response is instead

$$\langle 0 | E^* T_H(u) E | 0 \rangle = e^{-u/2}, \quad \int_0^\infty e^{-pu} \langle 0 | E^* T_H(u) E | 0 \rangle du = \frac{1}{p + 1/2}. \quad (\text{S12.18})$$

A finite polynomial endpoint gives only finitely many energies in this free control. An infinite response tower needs a specified observable or coupling and its matrix elements. A matching counting character does not supply them. This physical radial adjoint is also not an assertion that a tilded cohomological endpoint is the ordinary adjoint, or that (S12.9) has acquired a spatial spectral interpretation.

S12.6 Scope of the arithmetic comparison

The complete auxiliary arithmetic amplitude is

$$F_{\text{ar}}(p) = \pi^{-p/2} (p - \tfrac{1}{2}) \Gamma(p/2 + 5/4), \quad K_\omega^<(p) = \frac{F_{\text{ar}}(p - \omega)}{F_{\text{ar}}(p + \omega)}. \quad (\text{S12.19})$$

For the trial identification $s = (p + 1/2)/2$, the polynomial $N(2N - 1)$ acting on e^{-x} gives $(2x^2 - 3x)e^{-x}$ and Mellin factor $s(2s - 1)\Gamma(s)$. This matches the non-exponential part of (S12.19). It is an inverse fit, not the insertion selected by the separated scalar pair. Moreover the different trial offset $s = (p + 5/2)/2$ changes the required polynomial to $2N - 3$. A degree obstruction based on one trial map is therefore conditional, not universal.

The Mellin coordinate in (S12.4) is the magnitude of a boundary field. The spatial variable is logarithmic radius. The fixed-representative obstruction and the Hodge channel calculation provide no identification of these variables, their measures or their adjoints. Analytic continuation does not supply that missing intertwining relation, and a rescaling chosen to manufacture $\pi^{-p/2}$ does not explain the fixed arithmetic finite part. Nor can the coefficient of a Q -exact localizing deformation become the physical shift.

The useful physical alternatives must retain the unprotected angular modes, specify an enlarged or interface sector, or derive an observable coupling to the required composite states. They must then reproduce the causal response and its ordinary adjoint. The gamma amplitude established here explains a possible special-function structure; it supplies neither the complete arithmetic transfer nor an independent cumulative positive norm.

S13 Genuine trace growth in Yang–Mills theory

This section derives the expectation-value evolution summarized in the exposition. The contour is the actual Loewner prefix, not a reference contour deformed by an inverse map. We fix the theory and an observable before differentiating. The results are exact smooth-connection identities and their conditional quantum averages; no closed-form path integral is required.

S13.1 Theory, field resolution and averaging assumptions

Fix Euclidean $SU(N)$ Yang–Mills, $N \geq 2$, at $\theta = 0$. With Hermitian generators normalized by $\text{tr}(T^a T^b) = \delta^{ab}/2$, use

$$\begin{aligned} F_{\mu\nu}[A] &= \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu], \\ D_\mu^A X &= \partial_\mu X - i[A_\mu, X], \quad S[A] = \frac{1}{2g^2} \int_{\mathbb{R}^4} \text{tr} F_{\mu\nu}[A] F_{\mu\nu}[A] d^4x. \end{aligned} \quad (\text{S13.1})$$

The action, coupling and quantum state stay fixed. Embed the Loewner upper half-plane in $x_3 = x_4 = 0$; removing its hull does not change the physical spacetime or the measure.

To define a smooth probe, take the gauge-covariant field flow

$$\partial_\tau B_\mu = D_\nu^B F_{\nu\mu}[B], \quad B_0 = A, \quad \tau > 0 \text{ fixed}. \quad (\text{S13.2})$$

The Wilson observable is constructed from $B_\tau[A]$ and averaged in the original interacting measure. There is no independent Gaussian measure for B_τ . Field-flow time τ , Loewner capacity time t and the arithmetic shift ω are distinct. Positive-flow local observables and their perturbative renormalization are discussed in [31, 32].

Assumption S13.1 (Regularity for the expectation identities). The chosen regulated or flowed construction defines the indicated holonomies and insertion correlators on the swept compact region. Connections are sufficiently smooth for ordered-transport differentiation; expectations of the required local field bounds justify passing derivatives through the expectation. For the short-time remainder we require the two finite moments specified in Proposition S13.4.

This assumption is not a construction of nonperturbative continuum four-dimensional Yang–Mills. Results on local positive-flow observables alone do not establish it for every cusped nonlocal loop. Without positive flow the same algebra gives bare regulated identities, whose passage to renormalized unflowed observables must include perimeter, cusp and contact terms. We do not perform that passage here. Below we suppress τ and write A, F, D for the selected smooth connection; this does not identify B_τ with the integration variable in a Schwinger–Dyson equation.

S13.2 The actual prefix and its return chord

Use half-plane capacity $2t$ and a deterministic driver $u(t)$:

$$\partial_t g_t(z) = \frac{2}{g_t(z) - u(t)}, \quad q(t) = \lim_{y \downarrow 0} g_t^{-1}(u(t) + iy), \quad q(0) = 0. \quad (\text{S13.3})$$

We assume a trace that is C^1 for positive t , with the local integrability needed at zero. Section S13.5 supplies an explicit analytic example. No Brownian trace or contour average is assumed; general driver regularity is a separate issue [33].

Let U_t transport along $q[0, t]$, with later parameters ordered to the left. For $0 \leq r \leq 1$ let $R_t(r)$ transport from 0 to $rq(t)$ along the radial segment, and put

$$\mathbf{H}_t = R_t(1)^{-1} U_t, \quad W_\tau(t) = \frac{1}{N} \langle \text{tr} \mathbf{H}_t \rangle, \quad \widehat{F}_t(r) = R_t(r)^{-1} F_{12}(rq(t)) R_t(r). \quad (\text{S13.4})$$

The straight return chord closes the trace's color indices. It is a choice of observable, not an identity for every possible endpoint state. The trace is in the fundamental representation; basepoint conjugation leaves all the displayed traced insertions gauge invariant. At zero size $\mathbf{H}_0 = I$ and $W_\tau(0) = 1$.

Pure gauge transport reverses to its inverse. The same assertion does not follow for a scalar-arclength-coupled $\mathcal{N} = 4$ Wilson line: its scalar term requires an additional analysis. The chord-completed observable therefore supplies a pure-YM calculation within a fixed theory, not an implicit supersymmetric extension or a proposed all-input arithmetic transfer.

S13.3 Exact cancellation of the moving-tip term

Define

$$k_t = q_1 \dot{q}_2 - q_2 \dot{q}_1, \quad \mathbf{J}_t(r) = \int_0^r s \widehat{F}_t(s) ds, \quad \mathbf{J}_t = \mathbf{J}_t(1). \quad (\text{S13.5})$$

Proposition S13.2 (Growing-trace transport). *For a smooth Hermitian connection and the specified completion,*

$$\dot{\mathbf{H}}_t = ik_t \mathbf{J}_t \mathbf{H}_t. \quad (\text{S13.6})$$

Under Assumption S13.1,

$$\dot{W}_\tau = ik_t M_1, \quad M_1 = \frac{1}{N} \langle \text{tr}(\mathbf{J}_t \mathbf{H}_t) \rangle. \quad (\text{S13.7})$$

Proof. Only the newly added prefix segment contributes to

$$\dot{U}_t = iA_\mu(q(t)) \dot{q}^\mu U_t.$$

For the outward chord $x(s) = sq(t)$, its displacement is $s\dot{q}(t)$. The gauge part of the variation formula (S4.3) has $F_{\nu\mu} \dot{q}^\nu q^\mu = -k_t F_{12}$, and hence

$$R_t(r)^{-1} \dot{R}_t(r) = ir R_t(r)^{-1} A_\mu(rq) \dot{q}^\mu R_t(r) - ik_t \mathbf{J}_t(r). \quad (\text{S13.8})$$

At $r = 1$ the endpoint connection cancels exactly when $R_t(1)^{-1} U_t$ is differentiated. The remaining term is (S13.6). Tracing and averaging in the fixed measure gives (S13.7). No field equation, Gaussian contraction or factorization has been used. $\square$

For a planar curve with this chord, its oriented area satisfies

$$\mathcal{A}(t) = \frac{1}{2} \int_0^t k_v dv. \quad (\text{S13.9})$$

A constant commuting curvature gives $\mathbf{H}_t = \exp(iF_{12}\mathcal{A}(t))$, checking the orientation and the factor of two independently.

S13.4 Curvature transport and the next expectation equation

Differentiating the conjugation in (S13.4) with (S13.8) gives

$$\partial_t \widehat{F}_t(r) = r \dot{q}^\mu \widehat{D}_\mu \widehat{F}_{12_t}(r) + i k_t [J_t(r), \widehat{F}_t(r)]. \quad (\text{S13.10})$$

The endpoint connection in the conjugation converts the ordinary derivative of F into its covariant derivative. Define

$$\begin{aligned} D_{\mu,t} &= \int_0^1 r^2 \widehat{D}_\mu \widehat{F}_{12_t}(r) dr, & R_t &= \int_0^1 r [J_t(r), \widehat{F}_t(r)] dr, \\ P_t &= \int_{0 < s < r < 1} r s \widehat{F}_t(s) \widehat{F}_t(r) ds dr. \end{aligned} \quad (\text{S13.11})$$

In particular $J_t = \dot{q}^\mu D_{\mu,t} + i k_t R_t$. Splitting the square of its defining integral into the two triangles yields

$$J_t^2 + R_t = 2P_t. \quad (\text{S13.12})$$

Indeed, on $s < r$, the square contributes $rs(\widehat{F}_t(r)\widehat{F}_t(s) + \widehat{F}_t(s)\widehat{F}_t(r))$, and the commutator contributes their difference with the second ordering positive. Thus the smaller radial argument is on the left in P_t .

Proposition S13.3 (The first two quantum equations). *Under Assumption S13.1, let*

$$M_{D,\mu} = \frac{1}{N} \langle \text{tr}(D_{\mu,t} H_t) \rangle, \quad M_2 = \frac{1}{N} \langle \text{tr}(P_t H_t) \rangle.$$

Then, with $\mu = 1, 2$ summed,

$$\begin{aligned} \dot{W}_\tau &= i k_t M_1, & \dot{M}_1 &= \dot{q}^\mu M_{D,\mu} + 2i k_t M_2, \\ \ddot{W}_\tau &= i \dot{k}_t M_1 + i k_t \dot{q}^\mu M_{D,\mu} - 2k_t^2 M_2. \end{aligned} \quad (\text{S13.13})$$

Proof. The first equation is Proposition S13.2. Differentiate $J_t H_t$ and use (S13.12). Its derivative is $(\dot{q}^\mu D_{\mu,t} + 2i k_t P_t) H_t$. The second equation follows by tracing and averaging. The product rule gives the third on positive-time intervals with the required derivatives. $\square$

The equations retain the complete interacting expectation. The derivative and two-curvature insertions are new unknowns. Writing $\dot{W}_\tau/W_\tau$ as an unspecified coefficient is not a closure of the hierarchy.

S13.5 The explicit trace of a linear driver

Choose $u(t) = at$, $a \in \mathbb{R}$. At fixed initial z put $w_s = g_s(z) - as$, so $\dot{w}_s = 2/w_s - a$. For $a \neq 0$ a primitive of $w/(2 - aw)$ is

$$-\frac{w}{a} - \frac{2}{a^2} \log(2 - aw).$$

At the tip $w_0 = q(t)$ and $w_t = 0$. Integrating on the branch issuing into the upper half-plane gives

$$aq + 2 \log(1 - aq/2) = a^2 t, \quad \dot{q} = a - 2/q. \quad (\text{S13.14})$$

The local tip ODE is special to this driver; a general trace depends on the driver's history. In particular it is not obtained by substituting the singular tip into the forward map equation.

Set $s = \sqrt{t}$. The identity $qq_s = 2asq - 4s$ determines

$$q(t) = 2i\sqrt{t} + \frac{2a}{3}t - \frac{ia^2}{18}t^{3/2} + \frac{a^3}{135}t^2 + O_a(t^{5/2}). \quad (\text{S13.15})$$

For a direct analyticity argument, put $q = sv$ in (S13.14), divide out its double zero in s , and apply the implicit-function theorem at $v = 2i$; the derivative in v is nonzero. Therefore the coefficient recursion describes a local analytic series in s , not only a formal asymptotic ansatz. It follows that

$$\begin{aligned} \mathcal{A}(t) &= -\frac{2a}{9}t^{3/2} - \frac{11a^3}{1350}t^{5/2} + O_a(t^{7/2}), \\ k_t &= -\frac{2a}{3}\sqrt{t} - \frac{11a^3}{270}t^{3/2} + O_a(t^{5/2}). \end{aligned} \quad (\text{S13.16})$$

The negative sign for $a > 0$ records clockwise orientation. For $a = 0$, $q(t) = 2i\sqrt{t}$ and the return chord exactly retraces the prefix. Thus $\mathbf{H}_t = I$ for every field configuration. This is a control for the selected observable, not a universal statement about straight-line observables.

S13.6 A quantitative short-time expectation bound

For a fixed small terminal time t , let K_0 bound $\|F_{12}\|$ on all swept chords, and let K_1 bound $\|e^\mu D_\mu F_{12}\|$ for every unit planar vector e on that region. These bounds may depend on the field configuration. Put

$$I(t) = \int_0^t |k_v| dv, \quad I_1(t) = \int_0^t |k_v| |q(v)| dv, \quad F_0 = F_{12}(0). \quad (\text{S13.17})$$

Proposition S13.4 (Remainder with explicit constants). *For each smooth Hermitian connection,*

$$\left| \frac{1}{N} \text{tr } \mathbf{H}_t - 1 + \frac{\mathcal{A}(t)^2}{2N} \text{tr } F_0^2 \right| \leq \frac{K_0 K_1}{6} I(t) I_1(t) + \frac{K_0^3}{48} I(t)^3. \quad (\text{S13.18})$$

If $\langle K_0 K_1 \rangle$ and $\langle K_0^3 \rangle$ are finite on a fixed small swept neighborhood, the averaged inequality holds with these expected moments on the right. In particular, at fixed positive field-flow time and for the driver $u(t) = at$,

$$W_\tau(t) = 1 - \frac{2a^2}{81} C_\tau t^3 + O_{\tau,a}(t^{7/2}), \quad C_\tau = \frac{1}{N} \left\langle \text{tr } F_{12}[B_\tau](0)^2 \right\rangle. \quad (\text{S13.19})$$

Proof. Radial differentiation of the conjugated curvature gives

$$\left\| \widehat{F}_v(r) - F_0 \right\| \leq r |q(v)| K_1, \quad \|J_v - F_0/2\| \leq |q(v)| K_1/3.$$

Set $\mathbf{M} = \int_0^t k_v J_v dv$ and $S = K_0 I(t)/2$. Then

$$\|\mathbf{M} - \mathcal{A}(t) F_0\| \leq K_1 I_1(t)/3, \quad \|\mathbf{M}\| \leq S, \quad |\mathcal{A}(t)| \leq I(t)/2.$$

Each J_v is traceless, so the linear Dyson coefficient has zero trace. By cyclicity, the trace of the second ordered coefficient is $-\text{tr } \mathbf{M}^2/2$. The anti-Hermitian generator $ik_v J_v$ gives unitary evolution. The third-order integral remainder, with the remaining exact propagator retained, therefore has norm at most $S^3/6$; no exponential moment is needed. Finally,

$$\begin{aligned} \frac{1}{N} \left| \text{tr}(\mathbf{M}^2 - \mathcal{A}^2 F_0^2) \right| &\leq \|\mathbf{M} - \mathcal{A} F_0\| (\|\mathbf{M}\| + |\mathcal{A}| K_0) \\ &\leq K_0 K_1 I I_1/3. \end{aligned}$$

Dividing this contribution by two and adding $S^3/6$ proves (S13.18). Averaging uses the stated moments. For the linear driver, $I = O_a(t^{3/2})$ and $I_1 = O_a(t^2)$; (S13.16) supplies the coefficient in (S13.19). $\square$

No Wick expansion evaluates C_τ . It is a local moment of the same interacting theory and is nonnegative in a positive Euclidean measure. The order of limits is essential: $t \downarrow 0$ with τ fixed. Removing the resolution first can invalidate the moment bounds and introduces unflowed cusp and short-distance effects. This proposition is not a uniform smoothing-removal theorem or an unsmeared-loop OPE theorem.

S13.7 Schwinger–Dyson equations and the unresolved closure

At a finite lattice regulator, product Haar measure and the Wilson action $S_{\text{lat}} = \beta \sum_p (1 - \text{Re tr } U_p/N)$ give the exact integration-by-parts identity

$$\langle \nabla_e^a f \rangle = \langle f \nabla_e^a S_{\text{lat}} \rangle, \quad (\text{S13.20})$$

where ∇_e^a is a left-invariant link derivative. Applying it to Wilson observables gives deformation and splitting/joining terms. Finite- N expectations of products remain part of the system. Factorizing them is an additional large- N reduction, not a prerequisite for the quantum identity itself.

The continuum formal loop equation relates a covariant divergence of an area insertion to contact terms; its loop-space Laplacian form and supersymmetric extension are discussed in [34, Lecture 3]. Neither operator is automatically the first derivative along the specified Loewner family. The first difference can be exhibited at the field-component level. In unflowed notation set $E_\nu = \sum_{\mu=1}^4 D_\mu F_{\mu\nu}$. Then

$$\begin{aligned} \dot{q}^1 D_1 F_{12} + \dot{q}^2 D_2 F_{12} &= \dot{q}^1 E_2 - \dot{q}^2 E_1 \\ &\quad - \dot{q}^1 (D_3 F_{32} + D_4 F_{42}) + \dot{q}^2 (D_3 F_{31} + D_4 F_{41}). \end{aligned} \quad (\text{S13.21})$$

Consequently an equation for E_ν leaves an explicit transverse insertion in $\dot{q}^\mu M_{D,\mu}$: the second line, radially transported, integrated with weight r^2 , followed by H_t , traced and averaged. Planar placement of the contour does not delete it.

For a local algebraic control, take $T_3 = \sigma_3/2$ in $SU(2)$, $A_2 = b(x_1^2 - x_3^2)T_3/2$, and all other components zero. Then

$$F_{12} = bx_1 T_3, \quad F_{32} = -bx_3 T_3, \quad E_\nu = 0, \quad D_1 F_{12} = bT_3 = -D_3 F_{32}. \quad (\text{S13.22})$$

For $b = 1$ the deleted component would have operator norm $1/2$ despite the vanishing total equation-of-motion insertion. This is a counterexample to a component replacement, not a finite-action vacuum configuration or a no-go theorem for averaged closure. Any symmetry that removes its averaged counterpart must be proved for the chosen observable.

At positive flow time, $E[B_\tau]$ is not the functional derivative of $S[A]$. Applying (S13.20) differentiates $B_\tau[A]$ and produces its response kernel. The bare unflowed contact equation cannot be imported unchanged. The additional two-curvature moment, transverse insertion, flow-response kernel and possible unflowed cusp limit have not been reduced to W_τ and M_1 alone.

An enlarged system including loops and open lines has been derived for lattice Yang–Mills–Higgs [35]. That is methodological precedent with a different action, not a closure theorem for the fixed pure-YM observable here. The bounded next question is to evaluate the Schwinger–Dyson reduction of the specific derivative moment while retaining the displayed residuals. No closed expectation-value evolution, arithmetic realization or positive cumulative norm is claimed by the present hierarchy.

S14 Reproducible diagnostics and controls

The accompanying package contains four Python standard-library programs and their preserved JSON results. They register 299 finite cases. Each case records its name, numerical value, comparison direction, threshold and pass status. Replay requires identical parameters, case identities and thresholds and a passing result; small platform-dependent changes in floating values are permitted. No case uses zeta zeros or infers an arithmetic positivity theorem.

Program in <code>numerics/</code>	Cases	Main comparisons
<code>check_smooth_variation.py</code>	102	Inverse-map geometry, ordered ODE variation, endpoint motion, reparametrization, abelian and scalar-coupling controls.
<code>check_defect_endpoints.py</code>	99	Chiral projectors, all orientation signs, endpoint nullspaces, R charge and noncommuting adjoint transport.
<code>check_reflected_junction.py</code>	67	Reflected maps and color gluing, free Gram matrices, twist failures, smooth joins and independent bulk quadrature.
<code>check_fixed_window_response.py</code>	31	Full local/pole factor, cutoff finite part, finite-shift identities and unequal-radius Gaussian first variation.

For transparency the Clifford matrices may be generated by five tensor factors. With $j = 0, \dots, 4$ and Pauli matrices σ_i ,

$$\Gamma_{2j} = \sigma_3^{\otimes j} \otimes \sigma_1 \otimes I^{\otimes(4-j)}, \quad \Gamma_{2j+1} = \sigma_3^{\otimes j} \otimes \sigma_2 \otimes I^{\otimes(4-j)}. \quad (\text{S14.1})$$

They satisfy the stated Clifford relations. Projector traces are compared with independent row reduction. Correct triplet assignments and the allowed orientation supply nonzero controls for the excluded choices. The adjoint check uses noncommuting Hermitian matrix segments, so a mistaken reversal of matrix order cannot pass merely because an abelian example commutes.

The transport checks compare direct variation of an ordered ODE solution with the integrated insertion formula on smooth matrix backgrounds, including mesh refinement. The reflected response checks compare the analytic coefficient with a separately integrated double integral and check finite Gaussian current Gram matrices. The flat and ordinary-adjoint cases are positive or zero controls for the claimed failures. These comparisons support the written derivations; they do not certify physical supersymmetry transformations, quantum operator existence, continuum reflection positivity, all-order Ward identities or a stochastic contour limit.

The fixed-window program compares the subtracted arithmetic integral with the digamma symbol, checks the absorbed finite-shift pole factor and fixed contact term, and compares the Gaussian derivative with independent finite differences and the existing equal-radius control. These are binary64 diagnostics, not rigorous numerical enclosures. The boundedness and compactness obstructions are analytical proofs in Section S10; the auxiliary product in Section S11 is proved by finite-product algebra, not certified by the diagnostic count.

The standalone manuscript package can be checked with

```
python3 validation/drafts.py check -replay
```

The live investigation additionally has `validation/check_package.py`, which inventories the research notes and their inherited repository sources. That broader inventory is separate from the portable manuscript snapshot.

The auxiliary free-kernel background in Sections S1 and S2 is derived explicitly in the text. The 299 count refers to this investigation’s four programs, not to every calculation in its parent investigations. Hash and build records establish file identity and reproducibility, not mathematical correctness.

S14.1 Additional localization controls in version 0.5

Three further programs accompany Section S12. Their formats differ from the original case ledger, so no aggregate case count is assigned to them:

- `check_localization_hemisphere.py` compares paired modes, the complete propagator, Mellin moments, endpoint ordering, ordinary-adjoint norms and the explicitly conditional arithmetic ratio at two quadrature orders.
- `check_spatial_radial_descent.py` records 83 cases: 57 exact rational checks and 26 floating-point controls for the centered polarization, non-closure residual, operator ordering, spatial covariance and field-space transform.
- `check_hodge_radial_channel.py` checks 25 protected degree levels, 13 spatial one-particle levels, six covariance-channel comparisons and a rational partial-sum/integral-test enclosure.

Their sources and small records are included in the standalone version-0.5 snapshot. They can be replayed directly using each program’s `-output` option; the build guide gives commands. The first four programs remain the 299 cases replayed by `validation/drafts.py check -replay`. The additional controls audit the stated algebra and normalization; they do not verify the external supersymmetry input or certify arithmetic positivity. The separate EMA interval certificates are maintained in the critical-path investigation [27].

S14.2 Genuine trace-growth controls in the live revision

The separate NumPy program `check_growing_trace_hierarchy.py` accompanies Section S13. Its small record is `numerics/records/growing-trace-hierarchy-20260921.json`. It derives twelve tip-series coefficients by exact rational recursion and independently integrates the original prefix transport for $A_1 = T_1$, $A_2 = T_2$, then closes the chord. Four samples include positive, negative and zero driver slopes. Finite differences check the matrix and first-moment equations; a spatial gauge transformation checks the tip-frame cancellation. An Abelian constant-curvature comparison checks the area sign, and two deterministic samples evaluate the remainder inequality.

The largest recorded matrix-derivative discrepancy is below 1.314×10^{-10} ; the ordering identity differs by less than 2.259×10^{-16} . Doubling the integration resolution changes the selected loop by less than 4.523×10^{-13} . These are floating diagnostics of geometry and matrix calculus, not interval certificates, YM sampling, continuum renormalization checks or specialist review. They do not increase the historical 299-case ledger. Run the program separately with its `-output` option; the analytical remainder is Proposition S13.4.

S15 Provenance and contribution disclosure

This companion was separated from the unified Wilson–Loewner manuscript on 20 September 2026 by OpenAI GPT-6 (Codex), for Edward Baker. Version 0.5 shares the exposition’s version number to identify the matched pair. The configured reasoning effort was not exposed to the assistant in this session and is not inferred. Both documents are working research drafts without independent mathematical review.

Sections S1–S8 preserve the detailed arithmetic, free-field, contour, supersymmetry, endpoint and reflection calculations from versions 0.1–0.2. The three principal local source notes are *Smooth Wilson variation and the constant Loewner driver*, *Defect-compatible contour freedom and the endpoint pairing*, and *Physical reflection, the reference junction, and a first bulk response*, all dated 19 September 2026. The parent Wilson-lines work supplies the angular regulator and auxiliary Gaussian model; the Loewner and fractional-dimension investigations supply the transfer factorization and scoped realization restrictions. Section S9 transcribes the inherited cumulative continuation algebra [19] and now reports the separately certified fixed-window anchor and obstruction to the scalar append estimate [27, 28]. The subsequent internal operator-coupling bound is cited separately [29]; its proof stays in the critical-path investigation. The full program and hypotheses are in the exposition [1].

The original assembly checked the printed conventions for Loewner normalization, tangent and D3–D5 projectors, Baker endpoint transformations and semicircle, Dorn–Pershin reflection, and the Weil-form weights against the cited primary papers. This revision does not claim a new exhaustive literature or priority survey. No novelty is claimed for standard transport variation, tangent-coupled supersymmetry, the ray endpoint selection rule, or the classical Weil criterion.

Version 0.4 incorporates the fixed-window Wilson-response and localization notes of 20 September 2026. Sections S10–S11 add the full contact/pole comparison, the defined Gaussian readout and its scoped obstructions, and the auxiliary boson–fermion product. The localization direction follows Edward Baker’s proposal and his earlier open-line research. Its physical mode pairing and operator interpretation remain unproved. The additional literature checks concern Pestun’s fluctuation complex, Wang’s defect bilocal and determinant assumption, and the gamma-function identities. The 31 additional diagnostics bring the total to 299; this revision has not received independent mathematical review.

`MANUSCRIPT_SOURCES.json` records assembly and revision inputs. `BUILD_RECORD.json` binds both documents and supporting checks; each dated archive has a `SNAPSHOT.json`. These are identity records, not proof certificates. The historical unified drafts remain unchanged. Every TeX input needed to build either current document is in this package; no shared-background or parent-investigation TeX file is loaded at build time.

Version 0.5 adds Section S12, based on the predictive hemisphere, spatial radial-descent and Hodge-channel notes of 20 September 2026. The localization direction follows Edward Baker’s suggestion. The free boundary and Hodge inputs come from [25, 26]; the derived spatial restrictions do not exclude an enlarged or interacting sector. The three additional standard-library controls retain separate record formats and are not folded into the historical 299-case count. All new exposition and checks were prepared with OpenAI GPT-6 (Codex) assistance; the effort setting was not exposed and is not inferred. No specialist mathematical or physical review is claimed.

The live revision of 21 September 2026 incorporates Baker’s clarification that a specified Loewner evolution generates the curve defining the quantum observable, and that the immediate aim is its evolution equation in a fixed theory. The new chord-completed Yang–Mills calculation, its ordered hierarchy and quantitative remainder come from the research note [30]. They were derived and integrated with OpenAI GPT-6 (Codex) assistance; the reasoning effort was not exposed. The

field-theory regularity assumptions, finite matrix diagnostics and outstanding specialist review are stated separately. The archived version 0.5 and all earlier snapshots remain unchanged. At Baker's request this is an unarchived live revision: no new dated draft or version number has been saved. Arithmetic manuscript separation is deferred.

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